

Govt. of Karnataka

Department of Collegiate and Technical Education

GOVT. POLYTECHNIC BALLARI

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

IT SKILLS LAB (20CS01P)

CERTIFICATE

This is to certify that Mr/ Mrs
has satisfactory completed the course of experiments in Practical **IT SKILLS LAB (20CS01P)** prescribed by the Board of Technical Examination, Bengaluru for Sem Year Diploma in in this polytechnic during the year 20.... to 20....

INTERNAL ASSESSMENT MARKS

AVG OF 2 WRITTEN TESTS (20)	AVG OF 3 SKILL TESTS (20)	ACTIVITY (20)	TOTAL (60)

Signature of Staff

Signature of HOD

Examiners:

1.	2.
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Sl. No	Name of the experiment	Remarks
1	Write an algorithm for the following programmable problems • Add/subtract two numbers • Find the largest/smallest of 3 numbers • Calculate and print sum of 'N' numbers	
2	Design a flowchart for the following programmable problems • Add/subtract two numbers • Find the largest/smallest of 3 numbers • Calculate and print sum of 'N' numbers	
3	Design and create simple game using MIT-scratch / Code.org	
4	Design and create simple android application(MIT App Inventor)	
5	Design and create webpage for displaying your poem (Title, header, paragraph, formatting tags)	
6	Design and create webpage for your wish list (What you want to do). Also list challenges and opportunities along with images to present your dreams (List ordered and unordered, Image, table).	
7	Design and create webpage using HTML and CSS about an awesome animal (Use necessary CSS tags)	
8	Design and create web page for a travel book/recipe book with more than 3 pages, table to list places/recipes (iframe, hyperlink)	
9	Design and create web page with JavaScript to design a simple calculator to perform the following operations: sum, product, difference and quotient.	
10	Design and create a personal webpage with dashboard	
11	Design and create web page about advantages of business process	

IT SKILLS LAB MANUAL [20CS01P]

	automation with respect to your branch of engineering	
12	Create a workflow for education loan approval in bank/diploma admission process (Use any tool)	
13	Demonstrate ERP with ERPNext Demo for manufacturing, retail and service sector (Use any other ERP tools)	
14	Create user account and demonstrate use of Google drive, Google docs, Google Co-lab (Usage of Jupyter Notebook)	
15	Demonstrate Internet of Things using with examples a. Smart home b. Smart city c. Smart farming	
16	Installation of Antivirus software	
17	Demonstration and hands on browser settings	
18	Demonstration and hands on privacy settings and password policy	
19	Demonstration of common security threats (using videos) a. Phishing b. DoS attack c. Man in the middle attack d. Spamming e. Virus	

Program 01: Write an algorithm for the following programmable problems

- **Add/subtract two numbers**
- **Find the largest/smallest of 3 numbers**
- **Calculate and print sum of 'N' numbers**

Addition of two numbers:

Step 1: Start

Step 2: Read 2 numbers and store in num1 and num2.

Step 3: calculate the sum

$$\text{sum} = \text{num1} + \text{num2}$$

Step 4: Display sum.

Step 5: Stop

Subtraction of two numbers:

Step 1: Start

Step 2: Read 2 numbers and store in num1 and num2.

Step 3: calculate the difference

$$\text{sub} = \text{num1} - \text{num2}$$

Step 4: Display sub.

Step 5: Stop

Find the smallest of 3 numbers

Step 1: Start

Step 2: Read 3 numbers a, b, c.

Step 3: Compare a with b and c. If a is smaller than b and c then a is smallest among three numbers.

Step 4: Compare b with a and c. If b is smaller than a and c then b is smallest among three numbers.

Step 5: Compare c with a and b. If c is smaller than a and b then c is smallest among three numbers.

Step 6: Stop

Find the largest of 3 numbers

Step 1: Start

Step 2: Read 3 numbers a, b, c.

Step 3: Compare a with b and c. If a is larger than b and c then a is largest among three numbers.

Step 4: Compare b with a and c. If b is larger than a and c then b is largest among three numbers.

Step 5: Compare c with a and b. If c is larger than a and b then c is largest among three numbers

Step 6: Stop

Calculate and print sum of 'N' number

Step 1: Start

Step 2: Assign sum=0 and i=1

Step 3: Read n (limit)

Step 4: Repeat steps 5 to 6 until i=n reached

Step 5: Compute sum=sum+i

Step 6: Compute i=i+1

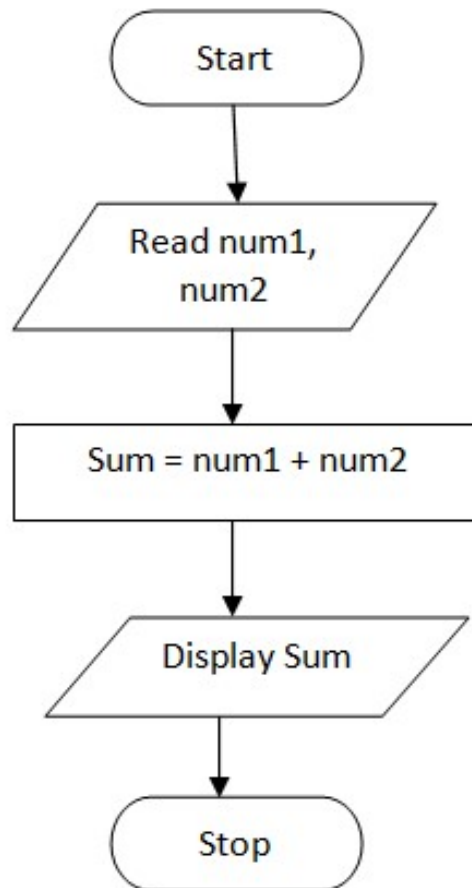
Step 7: Print sum

Step 8: Stop

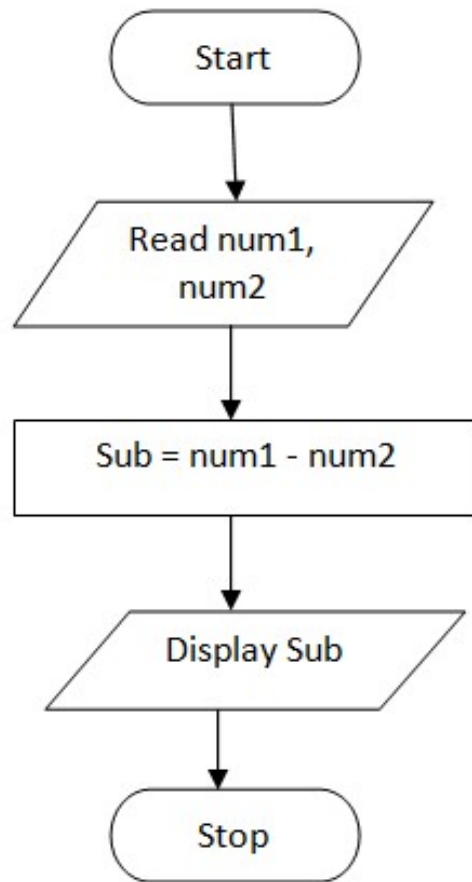
Program 02: Design a flowchart for the following programmable problems

- Add/subtract two numbers
- Find the largest/smallest of 3 numbers
- Calculate and print sum of 'N' numbers

Addition of two numbers:

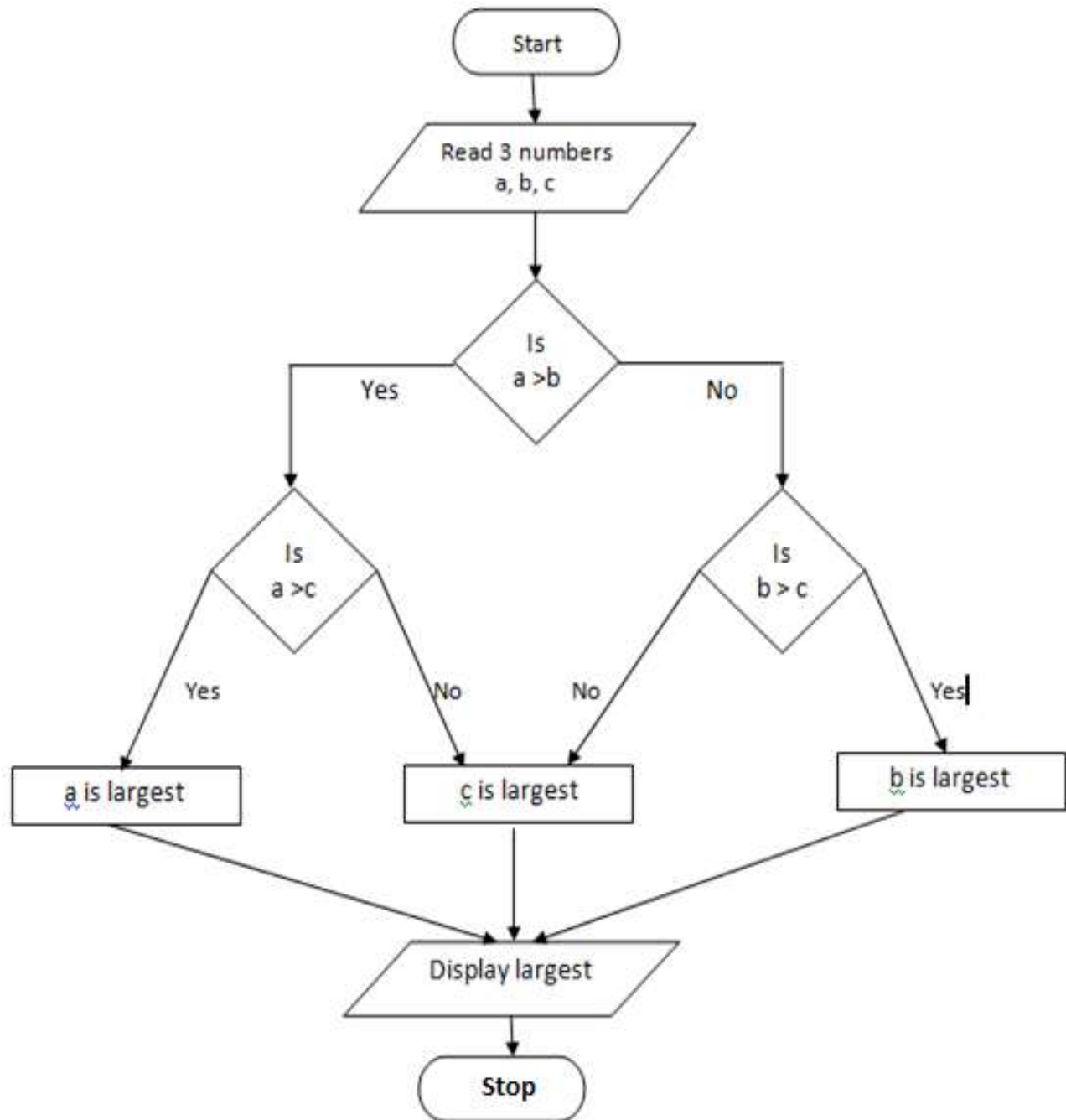


Subtraction of two numbers:

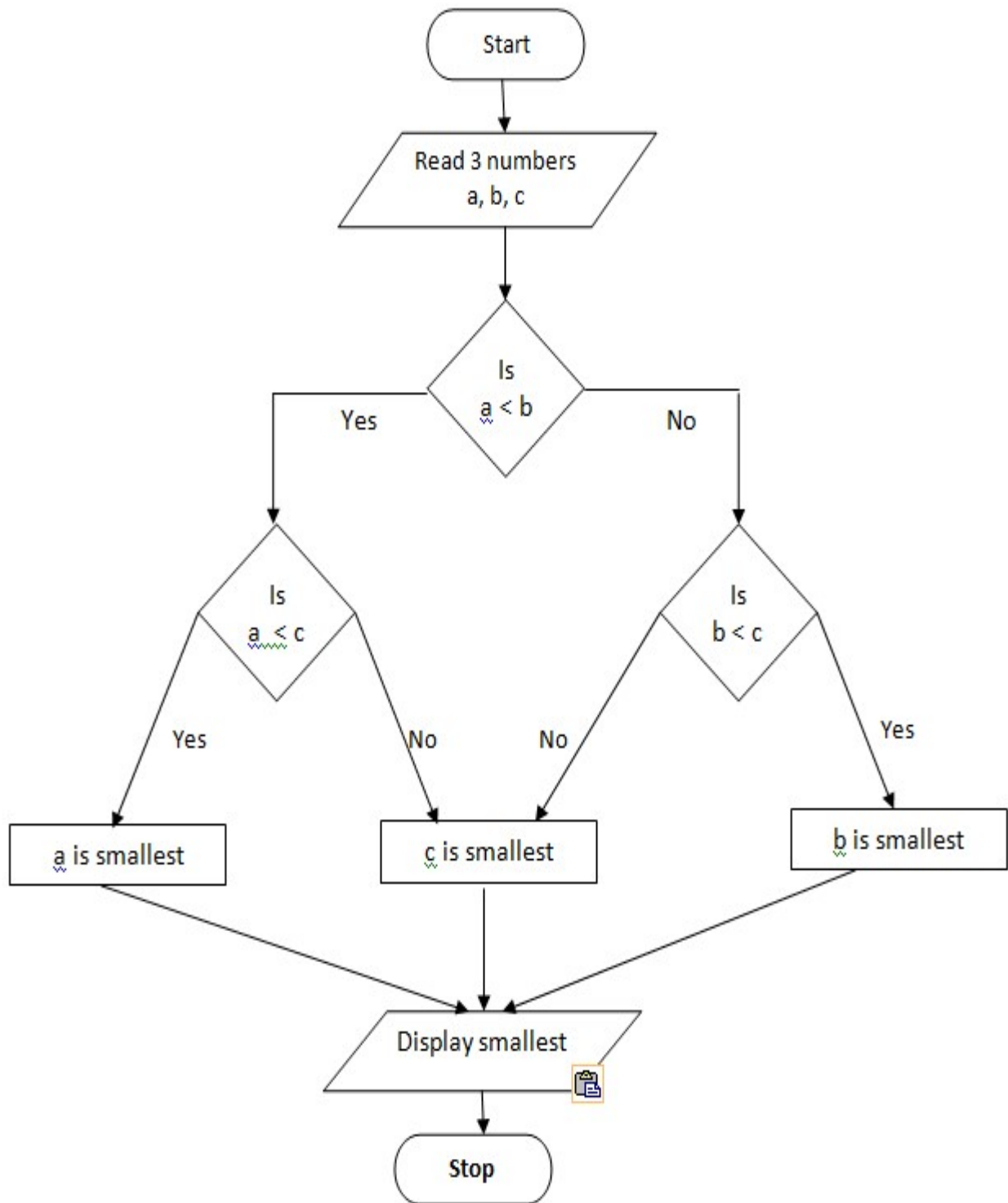


Find the largest/smallest of 3 numbers:

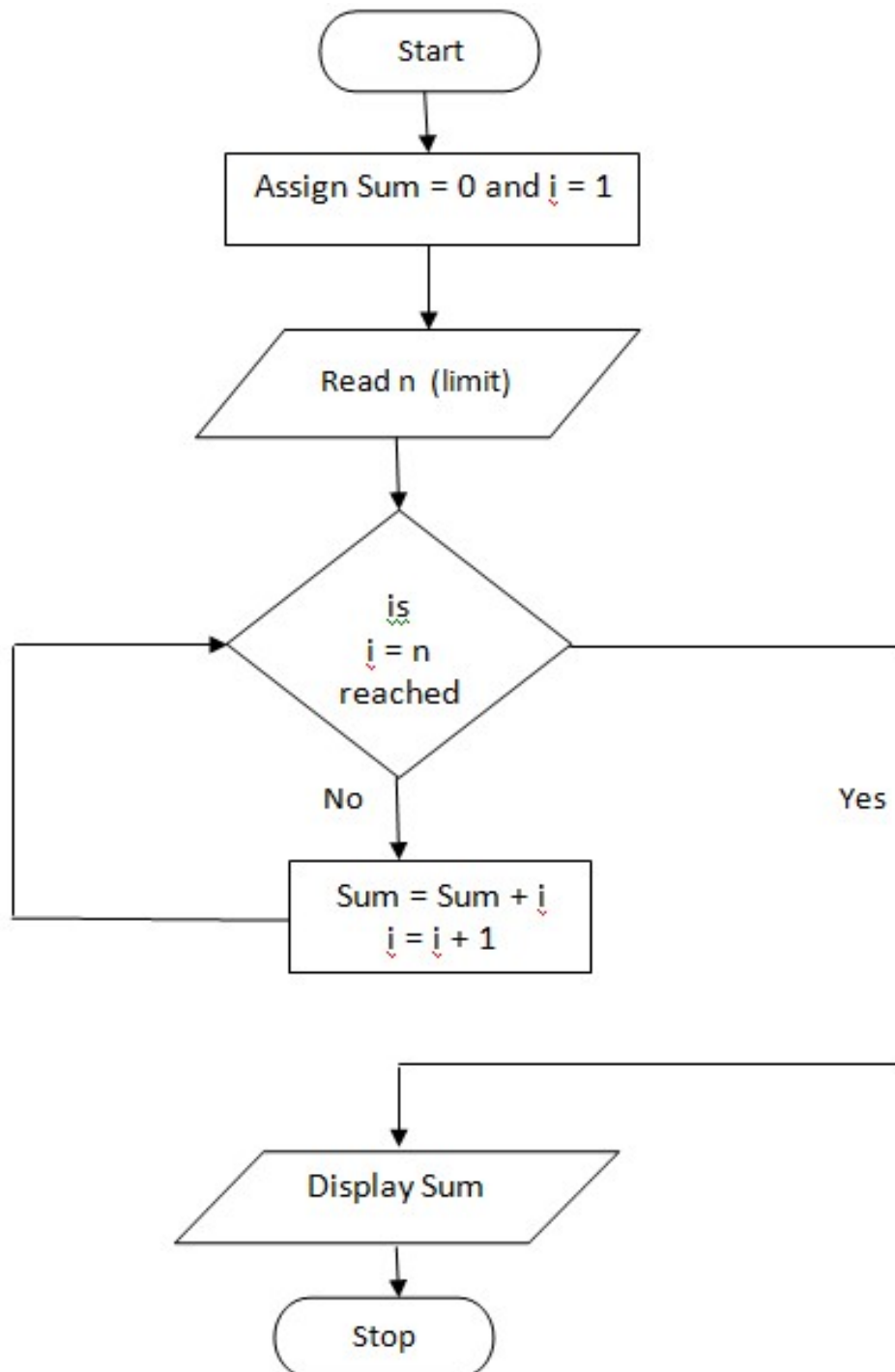
Largest of 3 numbers



Smallest of 3 numbers



Calculate and print sum of 'N' numbers:



Program 03: Design and create simple game using MIT-scratch/Code.org

Create a jumping game using scratch.

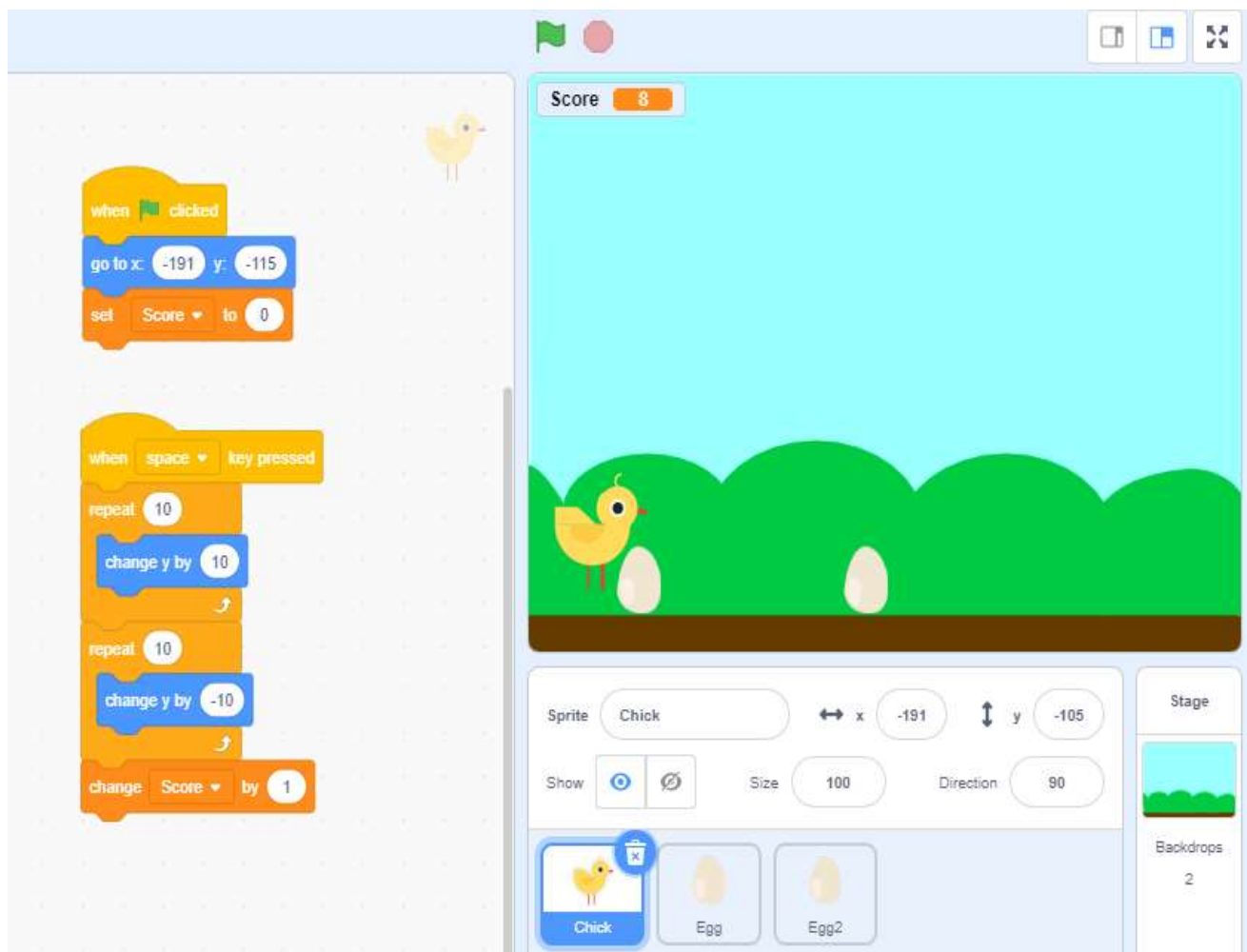
Steps:

1. Make your character to jump.
2. Make a moving obstacle.
3. Stop the game.
4. Add more obstacles.
5. Keep Score Procedure

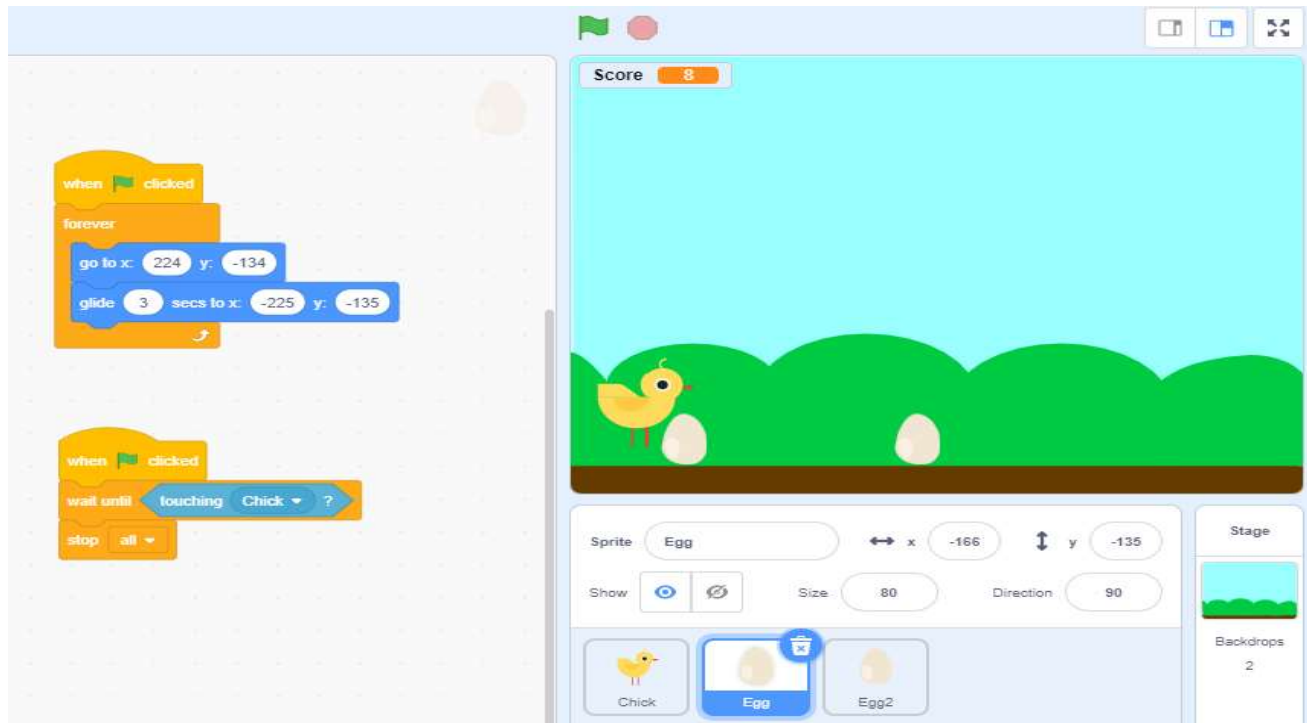
Watch this video and practice adding controls to code panel.

<https://www.youtube.com/watch?v=1jHvXakt1qw>

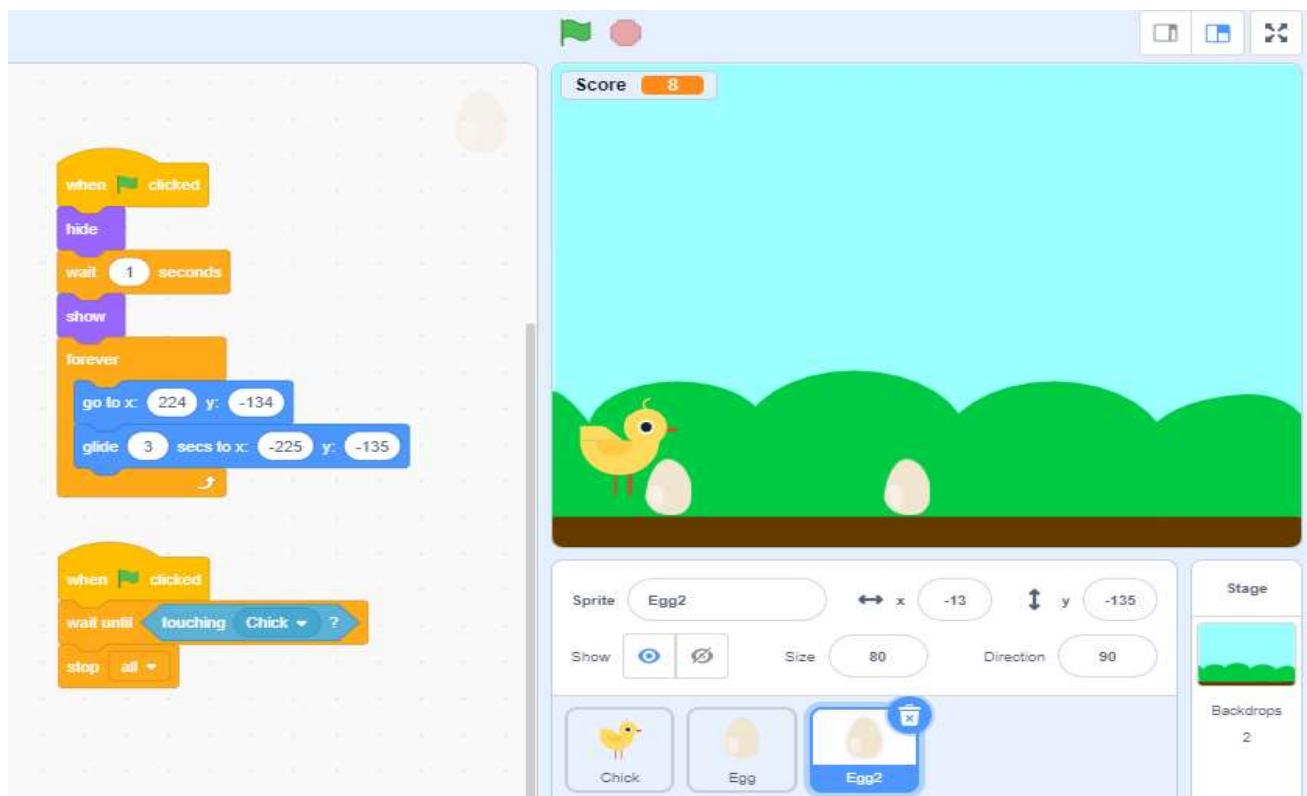
Select a chick from sprite and blue sky from backdrop. Add following script to chick.



Then add the following script to egg 1:



Then add the following script to egg 2:



Program 04: Design and create simple android application (MIT App Inventor)

STEPS:

- 1) In Google load MIT app inventor -> <https://appinventor.mit.edu/>
- 2) Login with your Gmail id
- 3) Install MIT AI2 Companion app in your mobile.
- 4) Then click on create an app on MIT app inventor website.



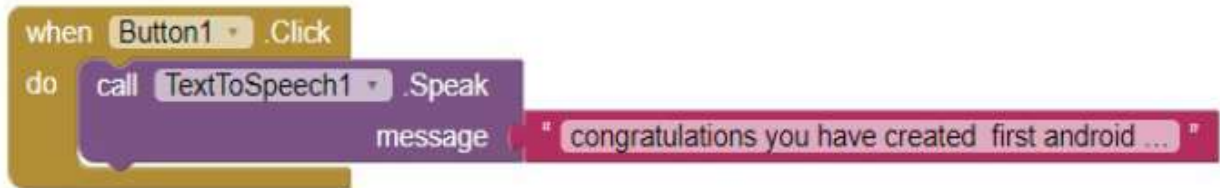
- 5) Click on projects → Start a new projects → Give name as FirstApp → Ok
- 6) Click on user interface then drag & drop a button on screen1
Change the text properties (right side down) as TalkToMe → Ok
- 7) Click on blocks (upper right corner) drag & drop button1.click



- 8) Click on designer (upper right corner) → Click on media (Left side options) → drag & drop text to speech on screen1 (It will be placed in components section)
- 9) Again Click on blocks → Go to text to speech1 → drag & drop call text to speech1.speak and place it inside Button1.click



10) Now click on blocks→text→drag &drop text block→then type text to speak msg
“congratulations you have created first android app”



11) Click on connect menu→AI companion→It generates QR code→ From mobile open MI inventor App then scan QR code.

12) Then run the app→reset connection→logout.

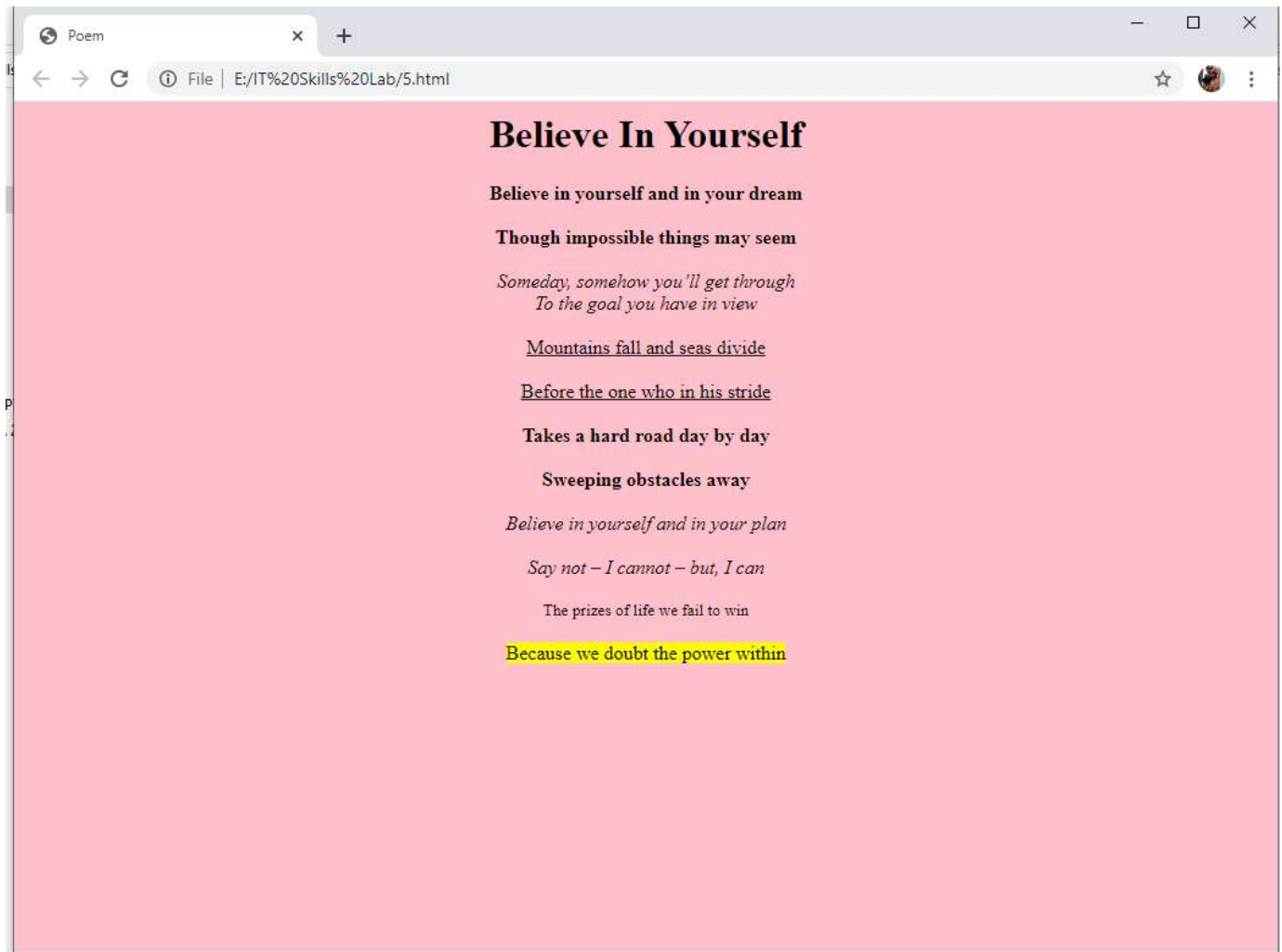
OUTPUT



Program 05: Design and create webpage for displaying your poem (Title, header, paragraph, formatting tags)

```
<html>
<head>
  <title>Poem</title>
</head>
<body bgcolor="pink">
  <center>
    <h1>Believe In Yourself</h1>
    <p>
      <b>Believe in yourself and in your dream <br><br>
      Though impossible things may seem </b> <br><br>
      <i>Someday, somehow you'll get through <br>
      To the goal you have in view </i> <br><br>
      <u>Mountains fall and seas divide <br><br>
      Before the one who in his stride </u> <br><br>
      <strong>Takes a hard road day by day <br><br>
      Sweeping obstacles away </strong><br><br>
      <em>Believe in yourself and in your plan <br><br>
      Say not – I cannot – but, I can </em><br><br>
      <small>The prizes of life we fail to win </small><br><br>
      <mark>Because we doubt the power within </mark><br><br>
    </p>
  </body>
</html>
```

OUTPUT

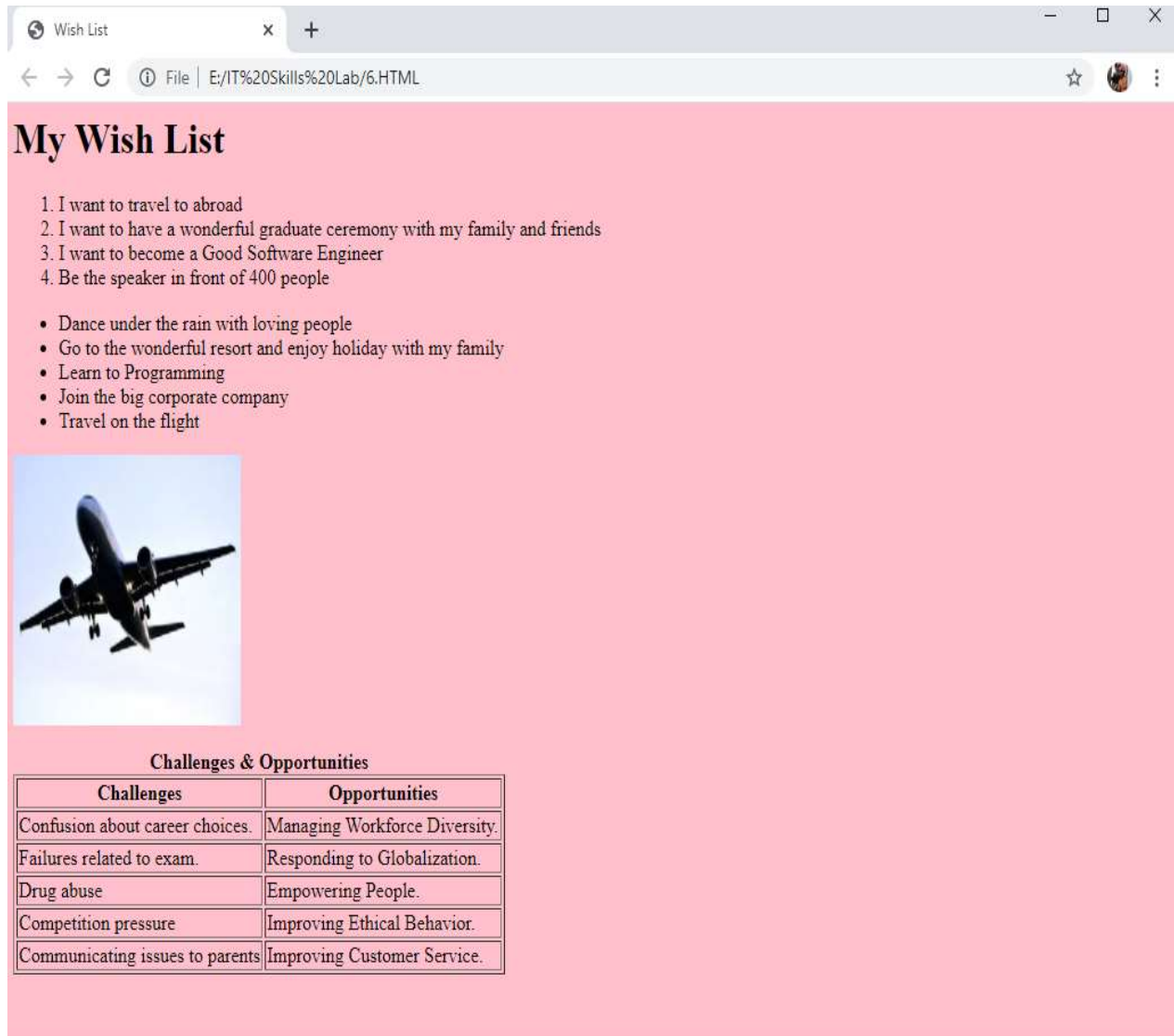


Program 06: Design and create webpage for your wish list (What you want to do). Also list challenges and opportunities along with images to present your dreams (List ordered and unordered, Image, table).

```
<html>
<head>
<title>Wish List</title>
</head>
<body bgcolor="pink">
<h1> My Wish List </h1>
<ol >
<li> I want to travel to abroad</li>
<li> I want to have a wonderful graduate ceremony with my family and friends </li>
<li> I want to become a Good Software Engineer</li>
<li> Be the speaker in front of 400 people</li>
</ol>
<ul>
<li> Dance under the rain with loving people</li>
<li> Go to the wonderful resort and enjoy holiday with my family</li>
<li> Learn to Programming </li>
<li> Join the big corporate company </li>
<li> Travel on the flight</li>
</ul>
</img> <br/><br/>
<table border=1>
<caption> <b>Challenges & Opportunities</b></caption>
<tr>
<th>Challenges</th>
<th>Opportunities</th>
</tr>
<tr>
<td> Confusion about career choices. </td>
<td> Managing Workforce Diversity.</td>
</tr>
<tr>
<td> Failures related to exam.</td>
<td> Responding to Globalization.</td>
</tr>
<tr>
<td> Drug abuse</td>
```

```
<td> Empowering People.</td>
</tr>
<tr>
<td> Competition pressure</td>
<td> Improving Ethical Behavior. </td>
</tr>
<tr>
<td> Communicating issues to parents</td>
<td> Improving Customer Service. </td>
</tr>
</table>
</body>
</html>
```

OUTPUT



My Wish List

1. I want to travel to abroad
2. I want to have a wonderful graduate ceremony with my family and friends
3. I want to become a Good Software Engineer
4. Be the speaker in front of 400 people

- Dance under the rain with loving people
- Go to the wonderful resort and enjoy holiday with my family
- Learn to Programming
- Join the big corporate company
- Travel on the flight



Challenges & Opportunities

Challenges	Opportunities
Confusion about career choices.	Managing Workforce Diversity.
Failures related to exam.	Responding to Globalization.
Drug abuse	Empowering People.
Competition pressure	Improving Ethical Behavior.
Communicating issues to parents	Improving Customer Service.

Program 07: Design and create webpage using HTML and CSS about an awesome animal (Use necessary CSS tags).

```
<html>
<head>
<title>Cascading Style Sheets(CSS)</title>
<style>
h1 {
color: red;
font-family: verdana;
font-size: 300%;
}
h2 {
color: blue;
font-family: Arial;
font-size: 200%;
}
p {
color: green;
font-family: courier;
font-size: 100%;
}
</style>
</head>
<body bgcolor="pink">
<h1>MY AWESOME ANIMAL IS ELEPHANT</h1>

<h2>characteristics of elephant</h2>
<p><b>
```

Elephants are the largest living land animal.They are characterized by its long trunk, columnar legs & huge head with temporal glands & wide, flat ears.

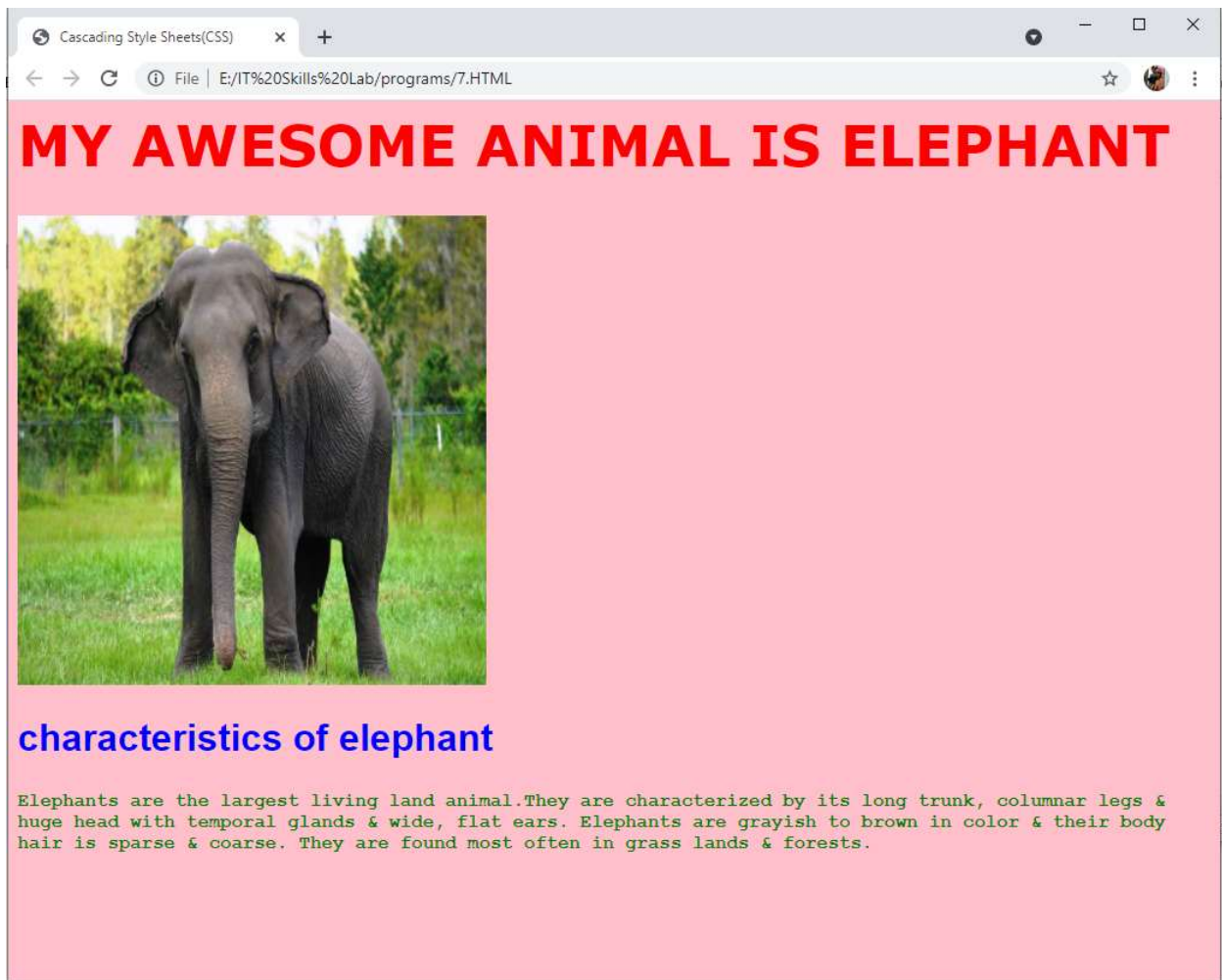
Elephants are grayish to brown in color & their body hair is sparse & coarse.

They are found most often in grass lands & forests.

</body>

</html>

OUTPUT



Program 08: Design and create web page for a travel book/recipe book with more than 3 pages, table to list places/recipes (iframe, hyperlink).

homepage.html

```
<html>
<head>
<title> Travel Book </title>
</head>
<body bgcolor="purple">
<center>
<font size="7" color="yellow"> MY TRAVEL BOOK WEBSITE </font>
<font size="4" color="yellow">
<marquee> Welcome to my Travel book website </marquee><br>
<iframe src="frame1.html" name="f1" style="width:30%; height:80%;"></iframe>
<iframe src="frame2.html" name="f2" style="width:65%; height:80%;"></iframe>
<marquee> Contact for more details: HOD, CSE, Government Polytechnic, Ballari </marquee>
</font>
</center>
</body>
</html>
```

frame1.html

```
<html>
<body bgcolor="lightpink">
<font size="6">
<center>
<table border=2 bordercolor="blue" cellpadding="10">
<tr>
<th bgcolor="lightgreen">TOURIST PLACES</th>
</tr>
<tr>
<td><a href="hampi.html" target="f2"> Hampi </a></td>
</tr>
<tr>
<td><a href="mysore.html" target="f2"> Mysore Palace </a></td>
</tr>
<tr>
<td><a href="bijapur.html" target="f2"> Bijapur Gumbaz</a></td>
</tr>
<tr>
<td><a href="badami.html" target="f2"> Badami Caves</a></td>
```

```
</tr>
<tr>
<td><a href="contact.html" target="f2"> Contact Details</a></td>
</tr>
</table>
</center>
</font>
</body>
</html>
```

frame2.html

```
<html>
<body bgcolor="lightpink">
</body>
</html>
```

hampi.html

```
<html>
<body>
<center>
<font color="white">
<h1>HAMPI</h1>
</font>

<table border="2" cellpadding="20">
<caption>
<font size="5" color="white">INFORMATION ABOUT HAMPI </font>
</caption>
<tr bgcolor="yellow">
<th>FOOD</th>
<th>HALT</th>
<th>GUIDE</th>
<th>TOURIST BUS</th>
</tr>
<tr bgcolor="lightyellow">
<td> Available </td>
<td> Available </td>
<td> Available </td>
<td> Available </td>
</tr>
```

```
</table>
```

```
</center>
```

```
</body>
```

```
</html>
```

mysore.html

```
<html>
```

```
<body>
```

```
<center>
```

```
<font color="white">
```

```
<h1>MYSORE</h1>
```

```
</font>
```

```

```

```
<table border="2" cellpadding="20">
```

```
<caption>
```

```
<font size="5" color="white">INFORMATION ABOUT MYSORE </font>
```

```
</caption>
```

```
<tr bgcolor="yellow">
```

```
<th>FOOD</th>
```

```
<th>HALT</th>
```

```
<th>GUIDE</th>
```

```
<th>TOURIST BUS</th>
```

```
</tr>
```

```
<tr bgcolor="lightyellow">
```

```
<td> Available </td>
```

```
<td> Available </td>
```

```
<td> Available </td>
```

```
<td> Available </td>
```

```
</tr>
```

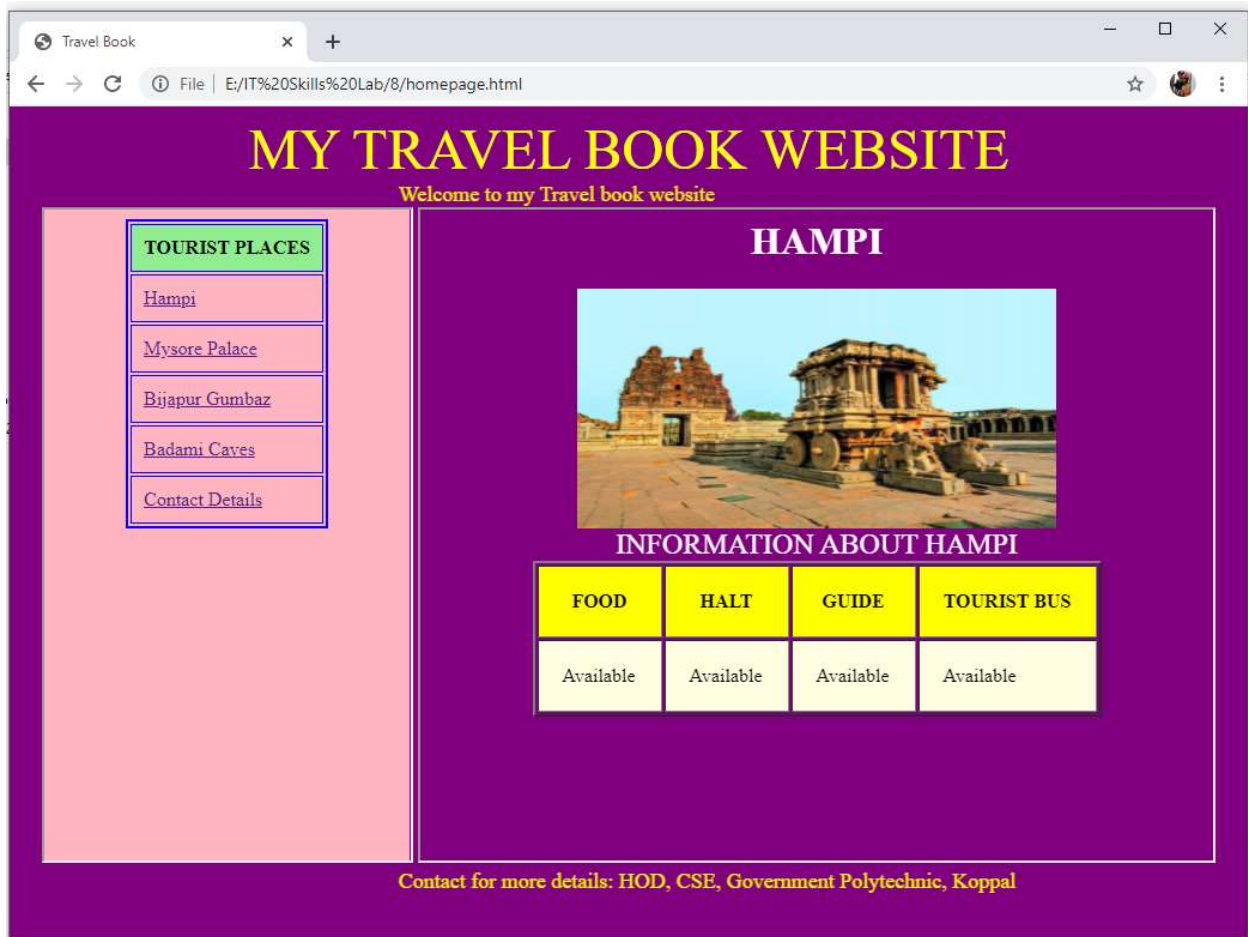
```
</table>
```

```
</center>
```

```
</body>
```

```
</html>
```

OUTPUT



Program 09: Design and create web page with JavaScript to design a simple calculator to perform the following operations: sum, product, difference and quotient.

```
<html>
<head>
  <title>Simple Calculator</title>
<script type=text/javascript>
function showresult(c)
{
  var n1=parseFloat(document.getElementById('num1').value);
  var n2=parseFloat(document.getElementById('num2').value);
  var r;
  switch(c)
  {
    case '1':
      r=n1+n2;
      break;
    case '2':
      r=n1-n2;
      break;
    case '3':
      r=n1*n2;
      break;
    case '4':
      r=n1/n2;
      break;
    default:
      break;
  }
  document.getElementById('result').value=r;
}
```

```
</script>
</head>
<body bgcolor="lightblue">
<center>
<h1 style="color:purple;">Simple Calculator</h1>
<table border="1" cellpadding="5" cellspacing="5" width="600">
<tr align="center">
<td style="color:green;"><b>First number</b></td>
<td style="color:green;"><b>Second Number</b></td>
<td style="color:red;"><b>Result</b></td>
</tr>
<tr align="center">
<td><input type="text" size=10 id='num1'></td>
<td><input type="text" size=10 id='num2'></td>
<td><input type="text" id='result' size=10 readonly ></td>
</tr>
<tr>
<td align="center" colspan="3">
<button style="color:blue;" onclick="showresult('1')"><b>SUM</b></button>
<button style="color:blue;" onclick="showresult('2')"><b>DIFFERENCE</b></button>
<button style="color:blue;" onclick="showresult('3')"><b>PRODUCT</b></button>
<button style="color:blue;" onclick="showresult('4')"><b>QUOTIENT</b></button>
</td>
</tr>
</table>
</body>
</html>
```

OUTPUT

Simple Calculator

First number	Second Number	Result
SUM DIFFERENCE PRODUCT QUOTIENT		

Program 10: Design and create a personal webpage with dashboard.

dashboard.html

```
<html>
<head>
<title>Personal Website Dashboard</title>
<style>
/* Style the header */
.header {
padding: 80px;
text-align: center;
background:tomato;
color: yellow;
}
/* Style the top navigation bar */
.navbar {
overflow:hidden;
background-color:yellow;
}
/* Style the navigation bar links */
.navbar a {
color: blue;
text-align: center;
padding: 14px 20px;
}
/* Change color on hover */
.navbar a:hover {
background-color:red;
color: black;
}
</style>
</head>
<body>
<div class="header">
<h1>My Website</h1>
<p>A website created by me</p>
</div>
<div class="navbar">
<a href="home.html">Home</a>
<a href="about.html">About</a>
<a href="contact.html">Contact Me</a>
```

```
</div>
</body>
</html>
```

home.html

```
<html>

<head>

<title>Home</title>

<body bgcolor="pink">

<p> Welcome to my dashboard website!</p>

</body>

</html>
```

about.html

```
<html>

<head>

<title>About</title>

<body bgcolor="yellow">

<p> I am first sem computer science student. I am studying in GPT Ballari</p>

</body>

</html>
```

contact.html

```
<html>

<head>

<title>Contact</title>

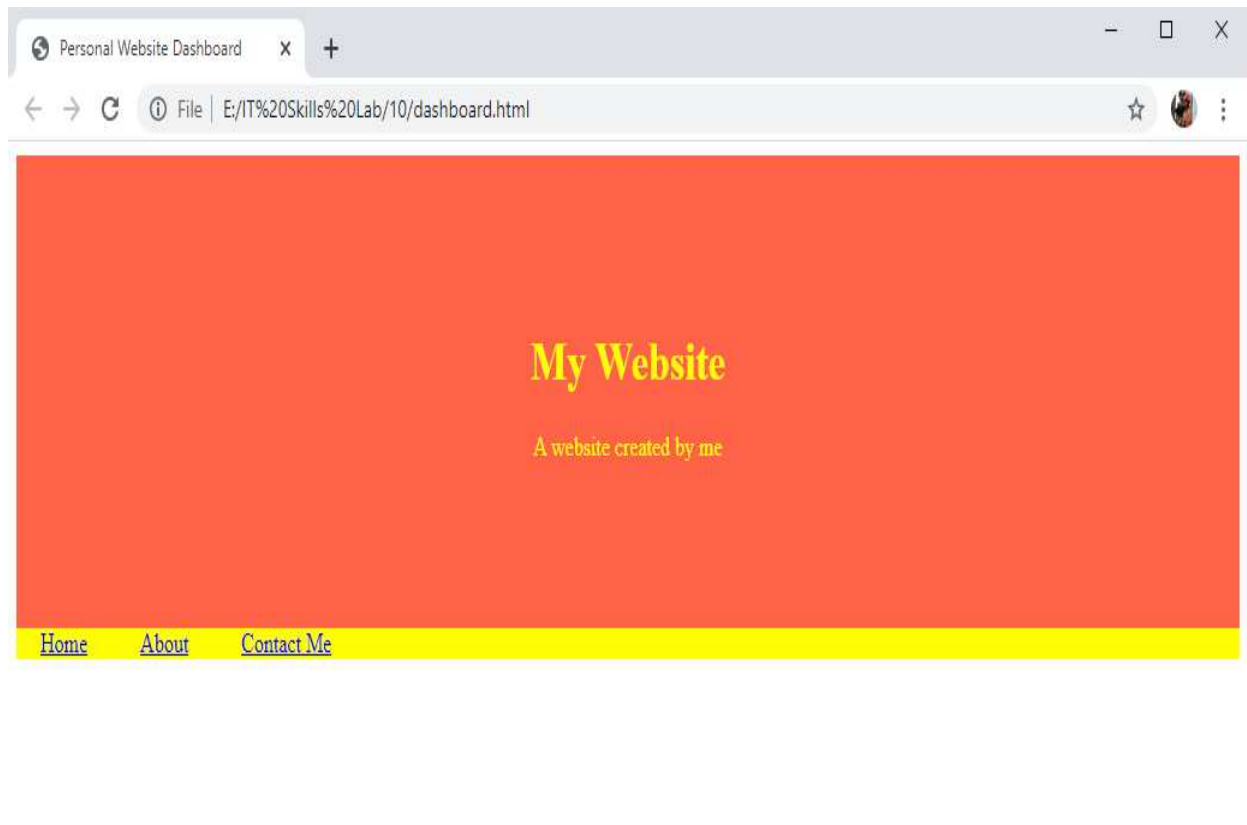
<body bgcolor="pink">

<p> Department of Computer Science & Engg. <br> Govt. Polytechnic Ballari</p>

</body>

</html>
```

OUTPUT



Program 11: Design and create web page about advantages of business process automation with respect to your branch of engineering

```
<html>
<head>
<title>Automation Testing Process</title>
</head>
<body bgcolor="lightblue" >
<h1 style="color:red;"> Advantages of Automation Testing in Computer Science and
Engineering</h1>
<ol>
<li>Automated testing takes far less resources in execution as compared to manual testing.</li> <br>
<li>Automated testing has less chances of error hence more reliable.</li> <br>
<li>Automated testing reduces the dependability of testing on the availability of the test
engineers.</li> <br>
<li>Automated testing provides round the clock coverage as automated tests can be run all time in
24*7 environment.</li> <br>
<li>Automated testing improves the coverage of testing as automated execution of test cases is faster
than manual execution</li> <br>
<li>It helps to train the test engineers to increase their knowledge by producing a repository of
different tests. </li> <br>
</img> <br><br>
</body>
</html>
```

OUTPUT

Automation Testing Process x +

File | D:/LAB%20MANUALS%202020/IT%20Skills%20Lab/programs/11.html

Advantages of Automation Testing in Computer Science and Engineering

1. Automated testing takes far less resources in execution as compared to manual testing.
2. Automated testing has less chances of error hence more reliable.
3. Automated testing reduces the dependability of testing on the availability of the test engineers.
4. Automated testing provides round the clock coverage as automated tests can be run all time in 24*7 environment.
5. Automated testing improves the coverage of testing as automated execution of test cases is faster than manual execution
6. It helps to train the test engineers to increase their knowledge by producing a repository of different tests.

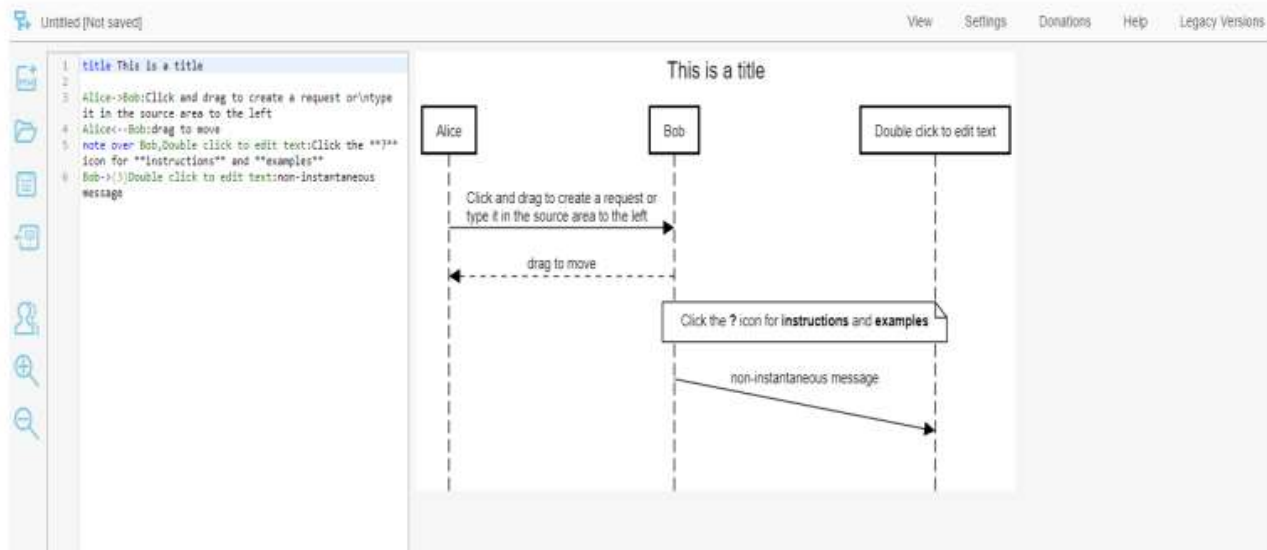
Automation Testing

- Fast
- Reliable
- Reusable
- Improves Accuracy
- Saves time and money
- Reduces Human-generated error
- Supports the execution of repeated test cases

Program 12: Create a workflow for education loan approval in bank/diploma admission process (Use any tool)

Visit the below web portal to create a sequence diagram for creating diploma admission process:

<https://sequencediagram.org/>



SEQUENCE

Title DIPLOMA ADMISSION PROCESS

STUDENT->COLLEGE: Applies for admission

STUDENT<--COLLEGE: Check SSLC/PUC marks and Provide application

STUDENT->COLLEGE: Fills in the details given in the application form along with the documents

STUDENT<--COLLEGE: Verifies the submitted documents

note over COLLEGE,DTE BOARD: **DTE provides a circular for documents verification**

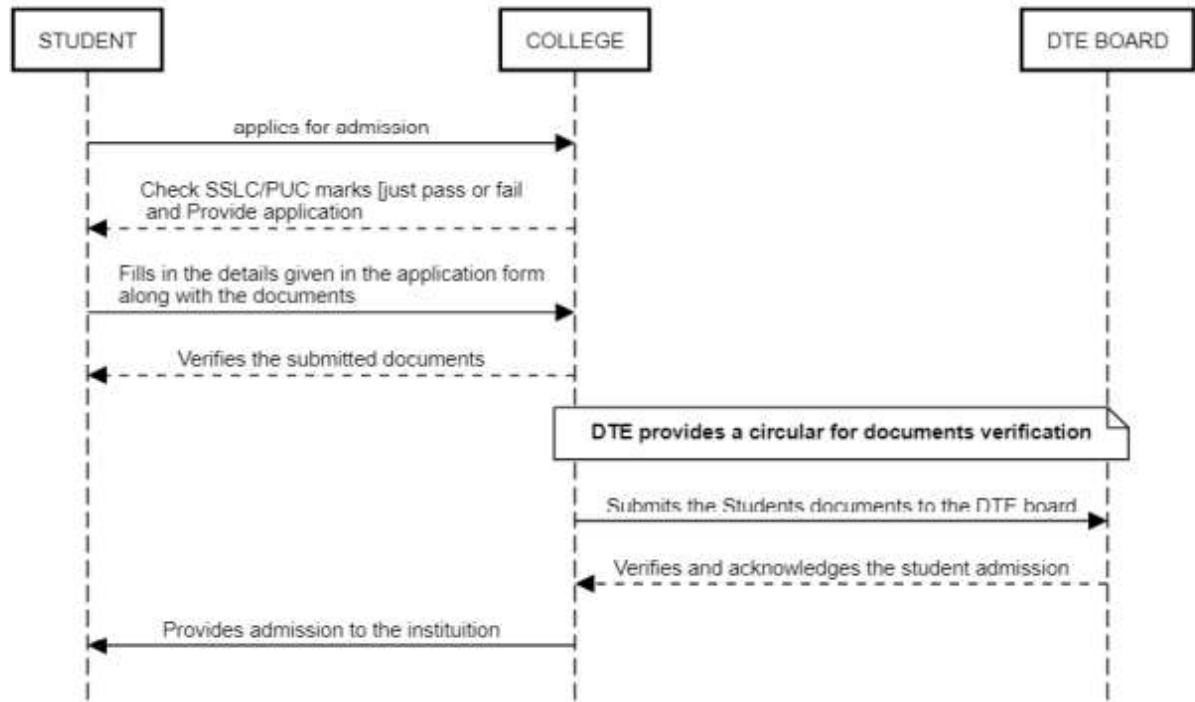
COLLEGE->DTE BOARD: Submits the Students documents to the DTE board

COLLEGE <-- DTE BOARD: Verifies and acknowledges the student admission

STUDENT <- COLLEGE: Provides admission to the institution

SEQUENCE DIAGRAM

DIPLOMA ADMISSION PROCESS



Program 13: Demonstrate ERP with ERPNext Demo for manufacturing, retail and service sector (Use any other ERP tools)

Click the below web portal and register your free account to maintain a Financial accounts

WEB PROTAL - <https://www.bitrix24.com/>

TUTORIAL VIDEO - <https://www.youtube.com/watch?v=ZQRuS6FmU68>

DEMO VIDEOS - <https://www.youtube.com/watch?v=yJ0c5jQ64mk>

<https://www.youtube.com/watch?v=kDa4279pj70>

Program 14: Create user account and demonstrate use of Google drive, Google docs, Google Co-lab (Usage of Jupyter Notebook)

CREATING G-MAIL ACCOUNT

Step 1: Go to www.gmail.com

Step 2: Click on create account.

Step 3: The Sign-up form will appear.

Step 4: Enter the required information.

Step 5: Click on “I agree” to the Google terms and services.

Step 6: Setup your recovery option.

Step 7: Now your account will be created and the welcome page will appear

HOW TO SIGN-IN

Step 1: Go to www.gmail.com

Step 2: Type Username & Password, and then click Sign-In.

HOW TO SIGN-OUT

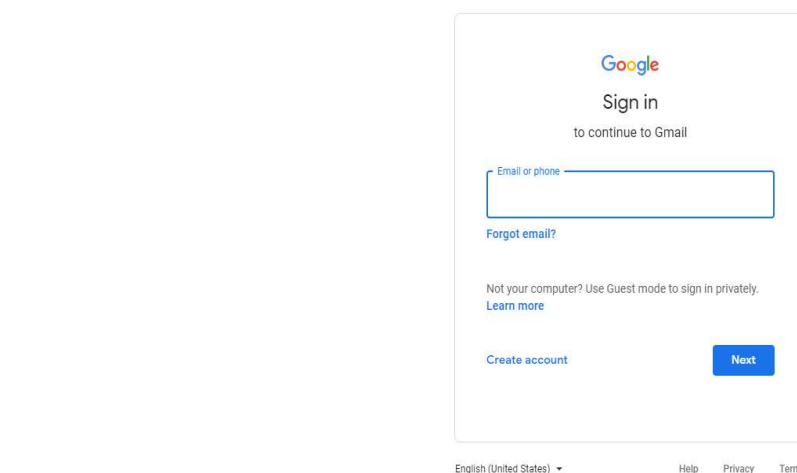
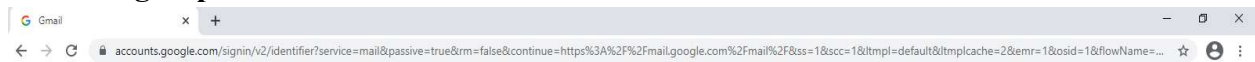
Step 1: In the top right corner of page, locate circle that has your first name initial.

Step 2: Click on the circle and select Sign-Out.

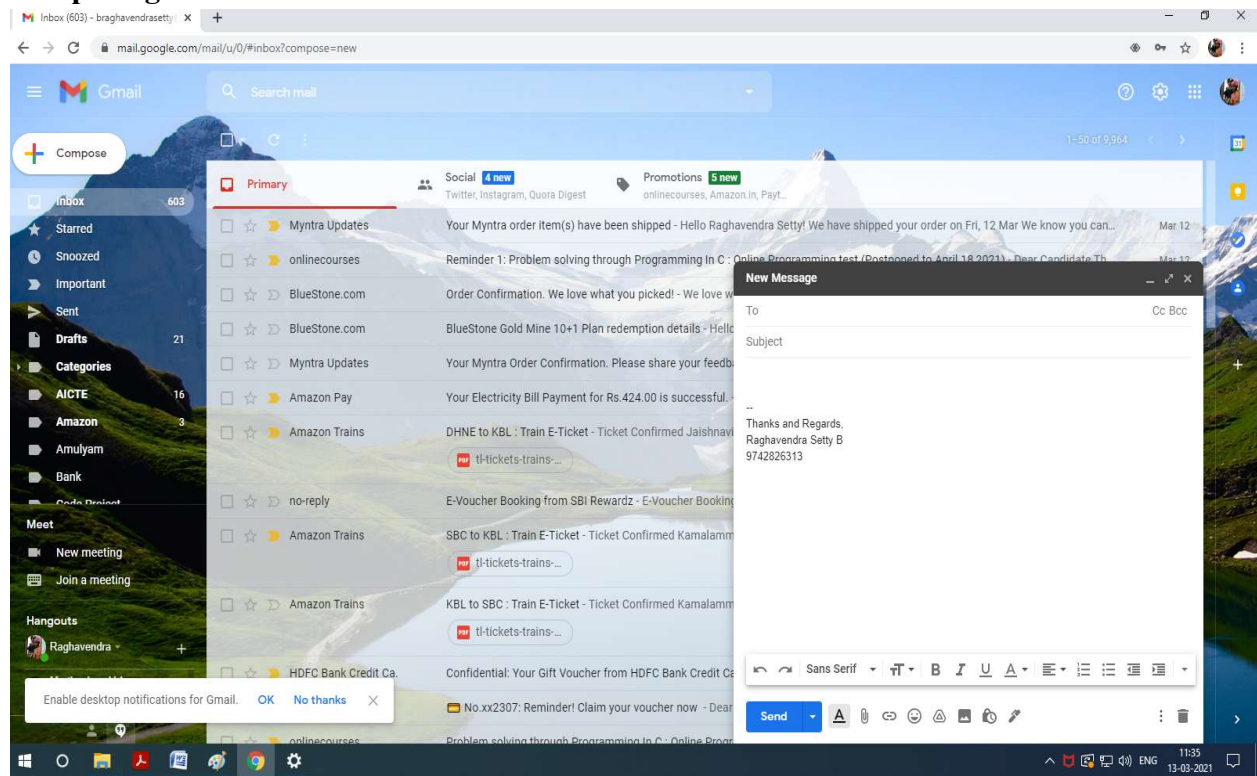
DEMONSTRATION LINKS (YOUTUBE VIDEOS):

1) Creating G-mail account: https://www.youtube.com/watch?v=_JFkz3WV4dQ

G-mail Sign-up form



Composing e-mail



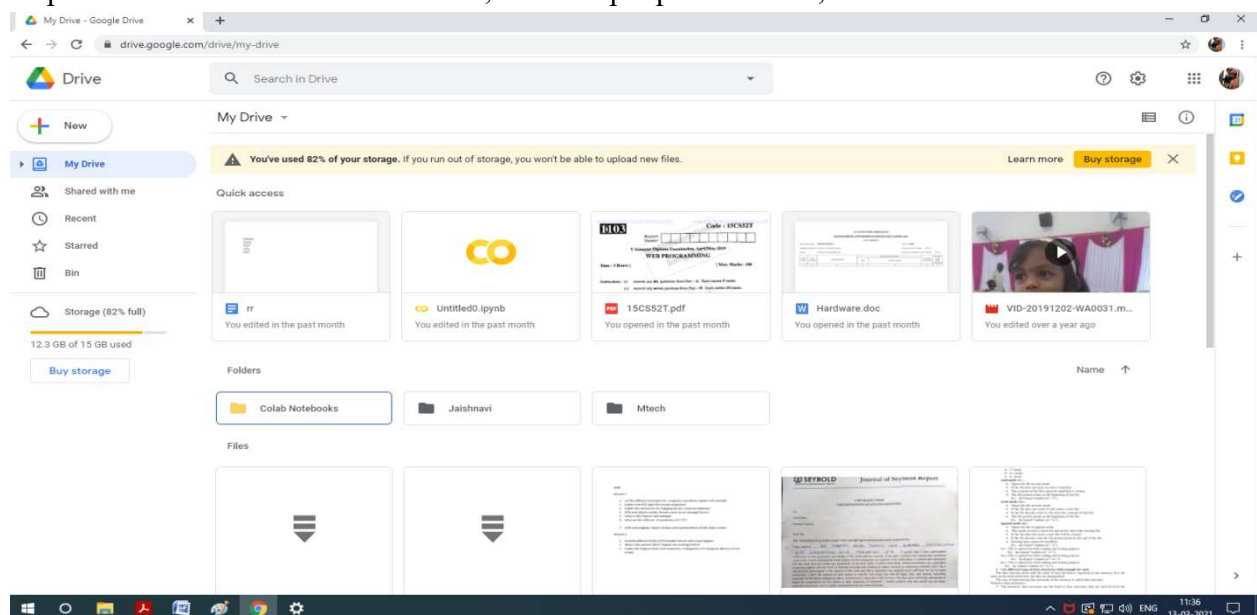
HOW TO USE “GOOGLE DRIVE”

Google drive can store our files securely and open or edit them from any device using Google drive. We get 15GB of space in our drive for free.

Step 1: Go to “drive.google.com”

Step 2: We can create files in Google drive or upload from our computers.

Step 3: We can share Files or Folders, so other people can view, edit on it.



DEMONSTRATION LINKS (YOUTUBE VIDEOS):

How to use Google Drive: <https://www.youtube.com/watch?v=82CUzggUxKU>

HOW TO USE “GOOGLE DOCS”

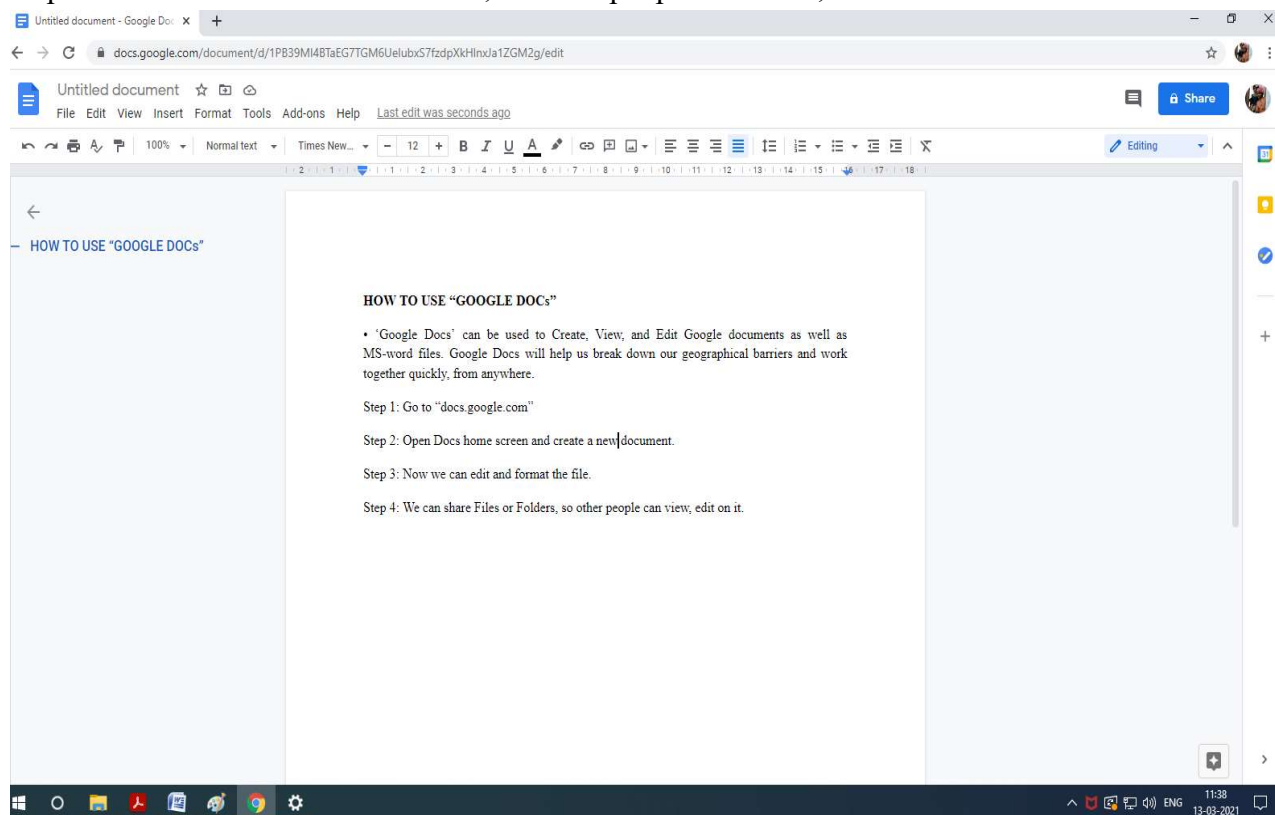
Google Docs can be used to Create, View, and Edit Google documents as well as MS-word files. Google Docs will help us break down our geographical barriers and work together quickly, from anywhere.

Step 1: Go to “docs.google.com”

Step 2: Open Docs home screen and create new document.

Step 3: Now we can edit and format the file.

Step 4: We can share Files or Folders, so other people can view, edit on it.



DEMONSTRATION LINKS (YOUTUBE VIDEOS):

How to use Google DOCS: <https://www.youtube.com/watch?v=xJiUTXGv3PE>

HOW TO USE “GOOGLE CO-LAB”

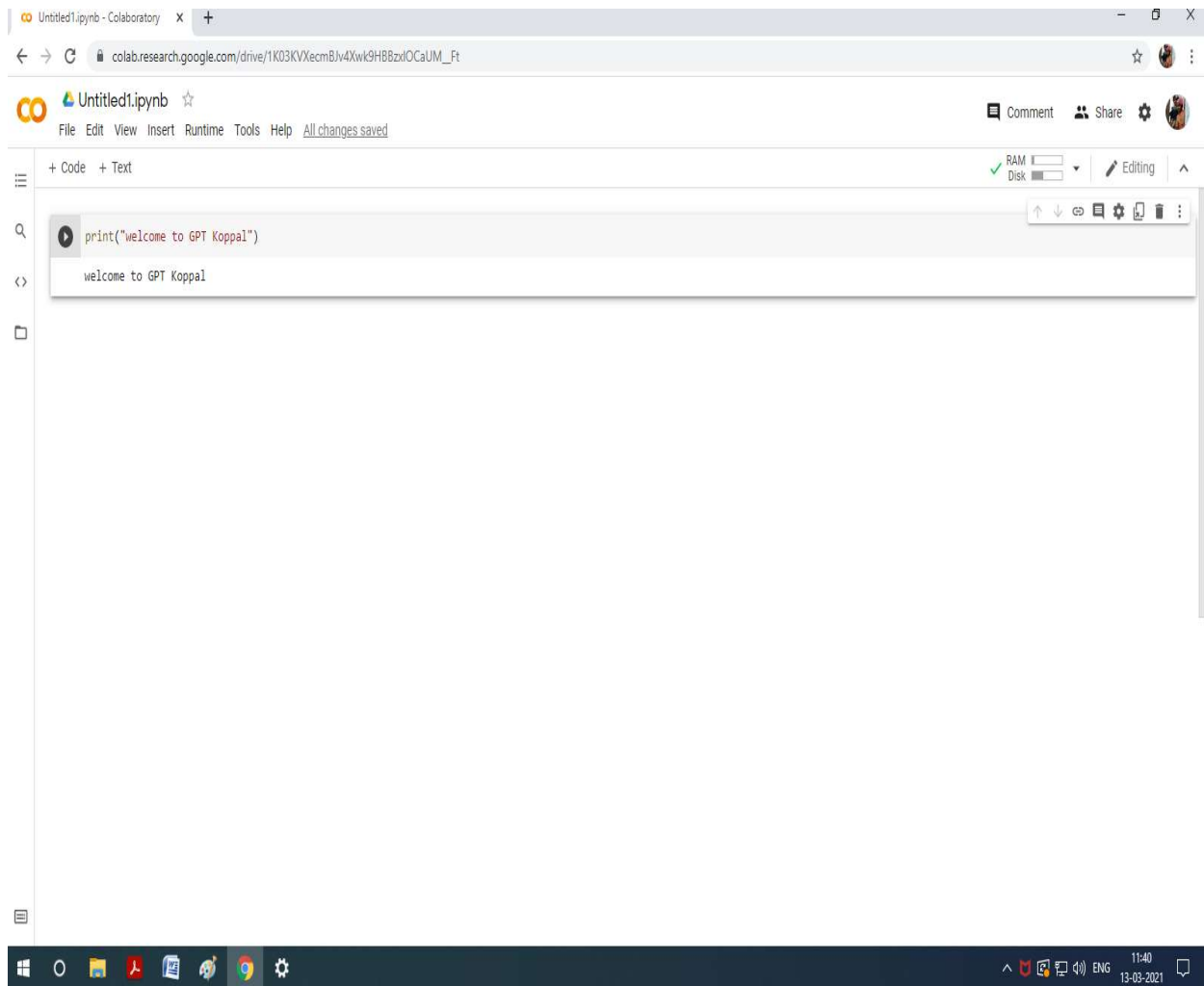
Google Collaborator (Google Co-lab) is a free online cloud-based Jupyter notebook environment that allows us to train our machine learning and deep learning models on CPUs, GPUs, and TPUs.

Step 1: Go to “colab.research.google.com”

Step 2: Open Colab home screen and create new Jupyter document.

Step 3: Now we can edit and run program file.

Step 4: We can share Files or Folders, so other people can view, edit on it.



DEMONSTRATION LINKS (YOUTUBE VIDEOS):

How to use Google Co-Lab: <https://www.youtube.com/watch?v=i-HnvsehuSw>

Program 15: Demonstrate Internet of Things using with examples

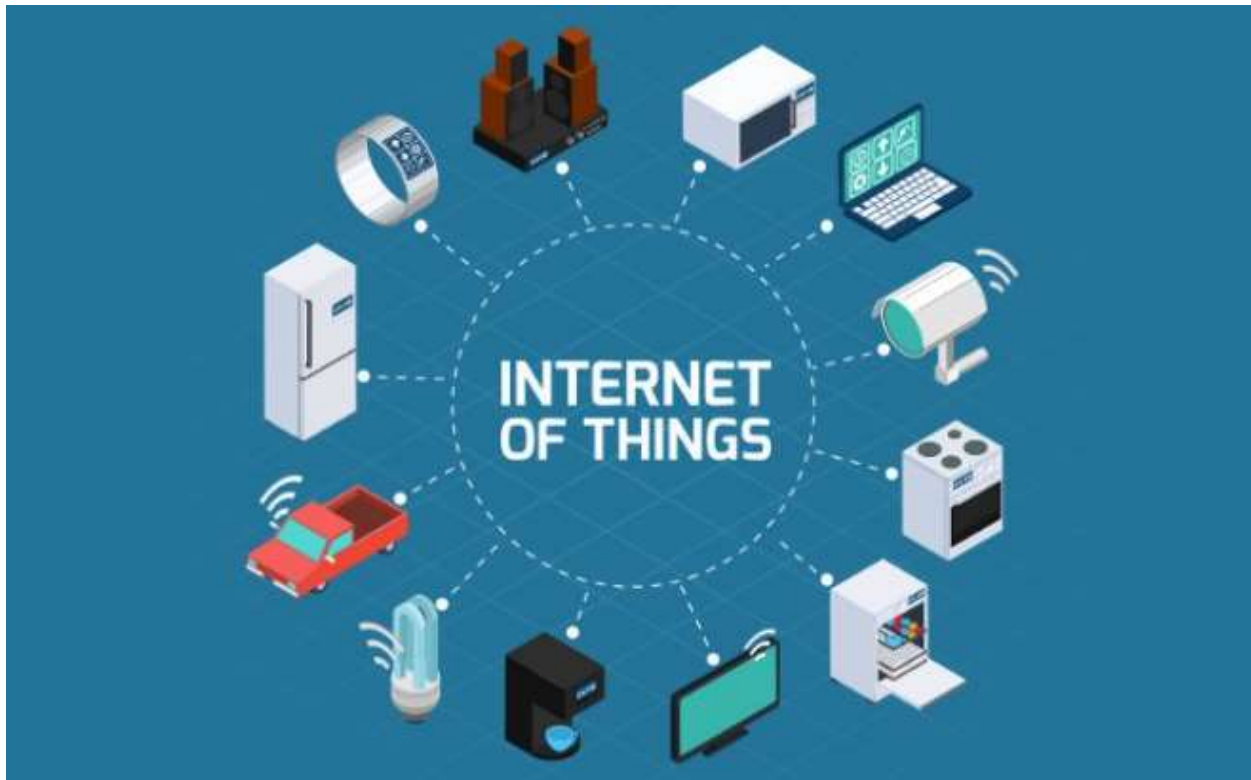
a. Smart home

b. Smart city

c. Smart farming

What is Internet of Things (IoT)

Internet of Things (IoT) refers to a system of interrelated, internet-connected objects that are able to collect and transfer data over a wireless network without human intervention. In simple words Internet of Things means taking all the things (devices) in the world and connecting them to the internet.



Internet of Things vs Internet

The primary difference between the Internet of Things and the Internet are the pillars for the concepts:

- The Internet of Things focuses on physical objects.
- The Internet o focuses on four constituents: people, things, data and processes.

Examples of “IoT” systems are

I) Smart Home

II) Smart City

III) Smart Farming etc.

I) SMART HOME: What Is a Smart Home?

- A smart home refers to a convenient home setup where appliances and devices can be automatically controlled remotely from anywhere with an internet connection using a mobile or other networked device.
- Devices in a smart home are interconnected through the internet, allowing the user to control functions such as security access to the home, temperature, lighting, and a home theater remotely.



How Smart Homes Work?

- A smart home's devices are connected with each other and can be accessed through one central point—a Smartphone, tablet, laptop, or game console.
- Door locks, televisions, thermostats, home monitors, cameras, lights, and even appliances such as the refrigerator can be controlled through one home automation system.
- The system is installed on a mobile or other networked device, and the user can create time schedules for certain changes to take effect.

Advantages of Smart Home system

- Installing a smart home technology system provides homeowners with convenience.
- Rather than controlling appliances, thermostats, lighting, and other features using different devices, homeowners can control them all using one device—usually a smart phone or tablet.
- Since they're connected to a portable device, users can get notifications and updates on issues in their homes.
- For the cost of setting up the smart system, homeowners can benefit from significant cost savings. Appliances and electronics can be used more efficiently, lowering energy costs.

Examples of Smart Home systems and devices.

- 1) Amazon Alexa
- 2) Google Assistant
- 3) Sony Smart TV

- 4) Perfect Bake pro
- 5) Amazon Echo etc.

II) SMART CITY

- A “Smart city” is a city that uses technology to provide services and solve city problems.
- A smart city does things like improve transportation and accessibility, improve social services, promote sustainability, and give its citizens a voice.
- The main purpose of the Smart Cities Mission is to drive economic growth and improve the quality of life of people by enabling local area development and harnessing technology, especially technology that leads to Smart outcomes.



Advantages of Smart Cities

- 1) Making more effective and data-based decisions:
- 2) Better transport services.
- 3) Efficient public services.
- 4) Reduction of the environmental footprint
- 5) Increase in digital equity
- 6) New economic development opportunities
- 7) Infrastructure improvement

Examples of Smart Cities in India

- 1) New Delhi
- 2) Mumbai
- 3) Bengaluru
- 4) Belagavi

- 5) Kolkata
- 6) Chennai
- 7) Hyderabad etc.

III) SMART FARMING (SMART AGRICULTURE)

- A “Smart Farming” is an emerging concept that refers to managing farms using modern Information and Communication Technologies to increase the quantity and quality of products while optimizing the human labor required.
- The goal of “Smart Farming” research is to ground a decision making support system for farm management. By providing them with the benefits of technological advancements, smart farming aims to reduce the heavy workload of the farm workers, hence improving their quality of life.



Advantages of Smart Farming

- 1) Increased Production.
- 2) Water Conservation.
- 3) Real-Time Data and Production Insight.
- 4) Lowered Operation Costs.
- 5) Increased Quality of Production.
- 6) Accurate Farm and Field Evaluation.
- 7) Improved Livestock Farming.

DEMONSTRATION LINKS (YOUTUBE VIDEOS):

- 1) Internet of Things (IoT): <https://www.youtube.com/watch?v=h0gWfVCSGQQ>
- 2) SMART HOME: <https://www.youtube.com/watch?v=EzCVpaEb0kQ>
- 3) START CITY: <https://www.youtube.com/watch?v=Br5aJa6MkBc>
- 4) START FARMING: <https://www.youtube.com/watch?v=DEqwIKwMM9k>

Program 16: Installation of Antivirus software

To install an antivirus program on your computer, follow the steps below.

Step 1: If you purchased the antivirus program from a retail store, insert the CD or DVD into the computer's disc drive.

Step 2: If you downloaded the antivirus program on the Internet, find the downloaded file on your computer.

Step 3: Unzip the file to extract and access the installation files. Double click on setup.exe or install.exe file.

Step 4: On the first screen you have to choose the type of installation:

- Regular installation • Custom installation

Step 5: Read and Accept End User License Agreement.

Step 6: Then, click Install to proceed with default installation

Step 7: Wait while setup installs Antivirus on your PC

Step 8: Once the installation is complete, click Continue

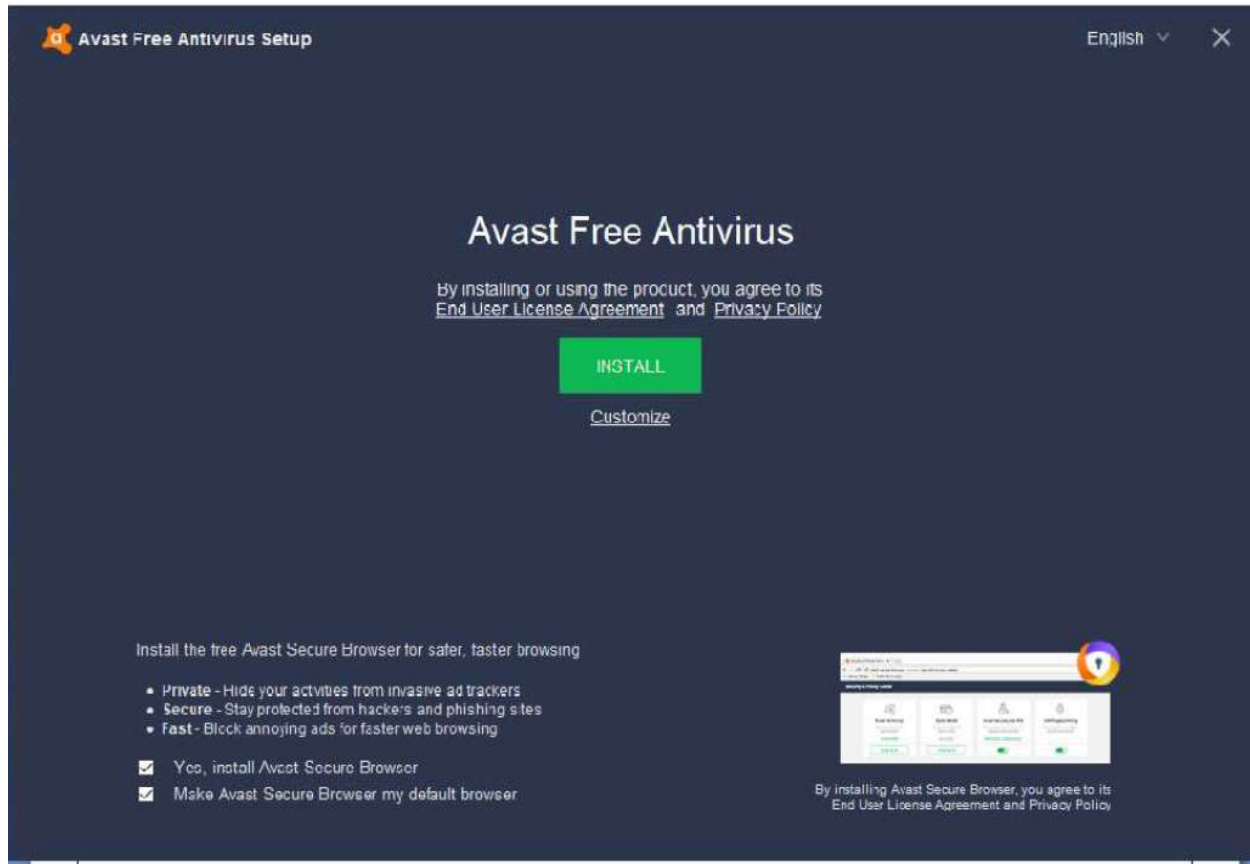
Step 9: Click on Scan to start scanning and to detect viruses, malware, and bad browser add-ons, and other issues on your PC.

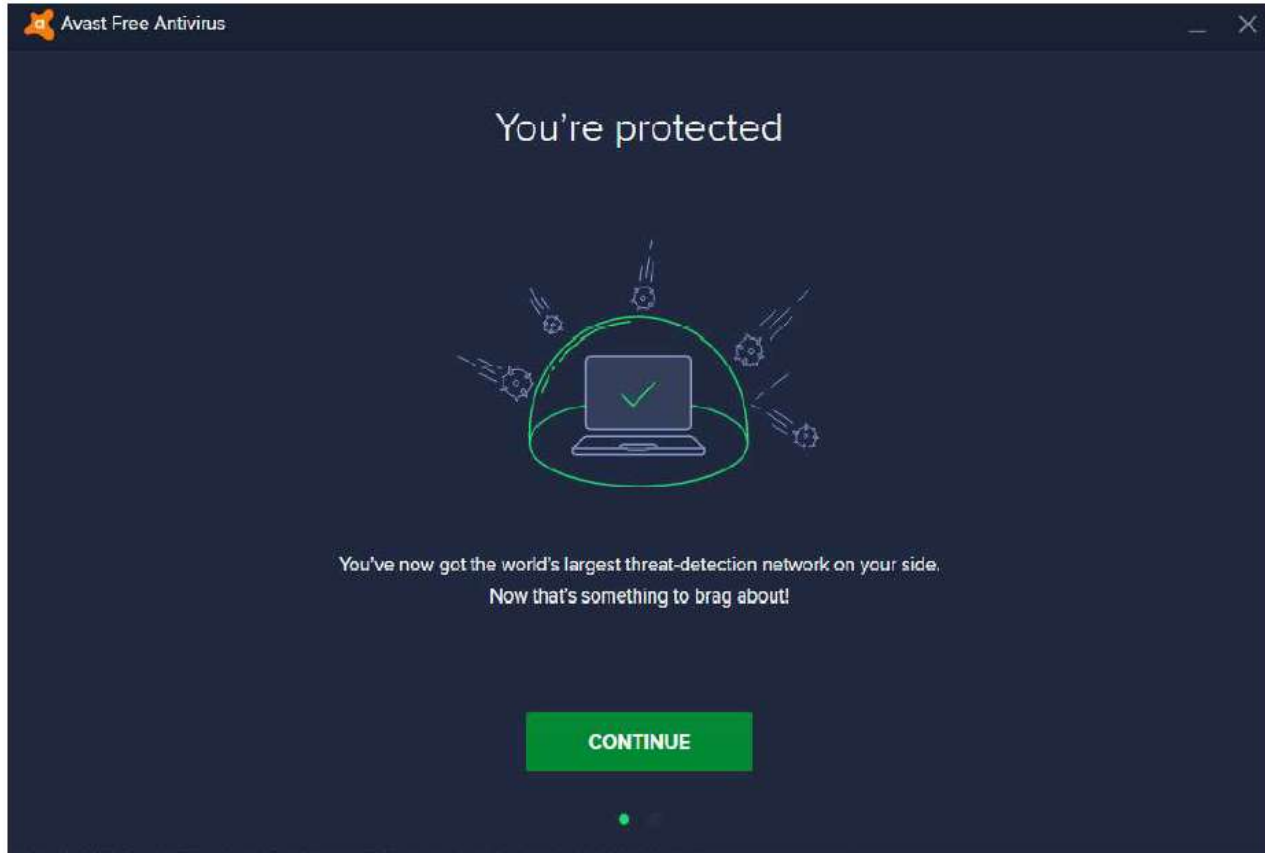
DEMONSTRATION LINKS (YOUTUBE VIDEOS):

Installation of Antivirus: <https://www.youtube.com/watch?v=rk0WcOT-Z2Y>

Free “Avast” Antivirus installation







Program 17: Demonstration and hands on browser settings.

GOOGLE CHROME BROWSER SETTINGS

Step 1: Open Google chrome web browser.

Step 2: Click Menu bar in the top right corner of the screen.

Step 3: Select settings from the dropdown menu.

Step 4: Here different settings options of browser appears (Ex. Appearance, Privacy & Security, Search Engine etc...)

Step 5: Click on respective options to make necessary changes.

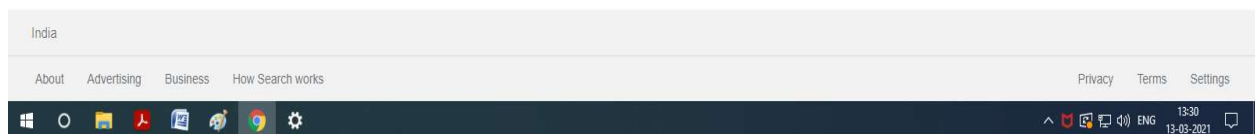
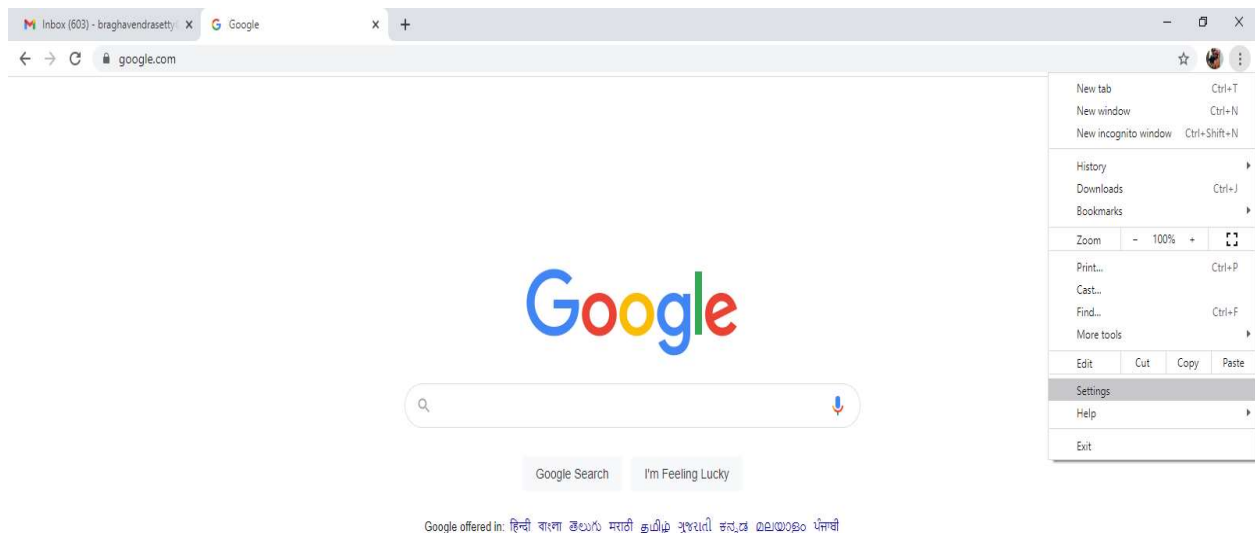
Step 6: Ex. Click on Appearance settings.

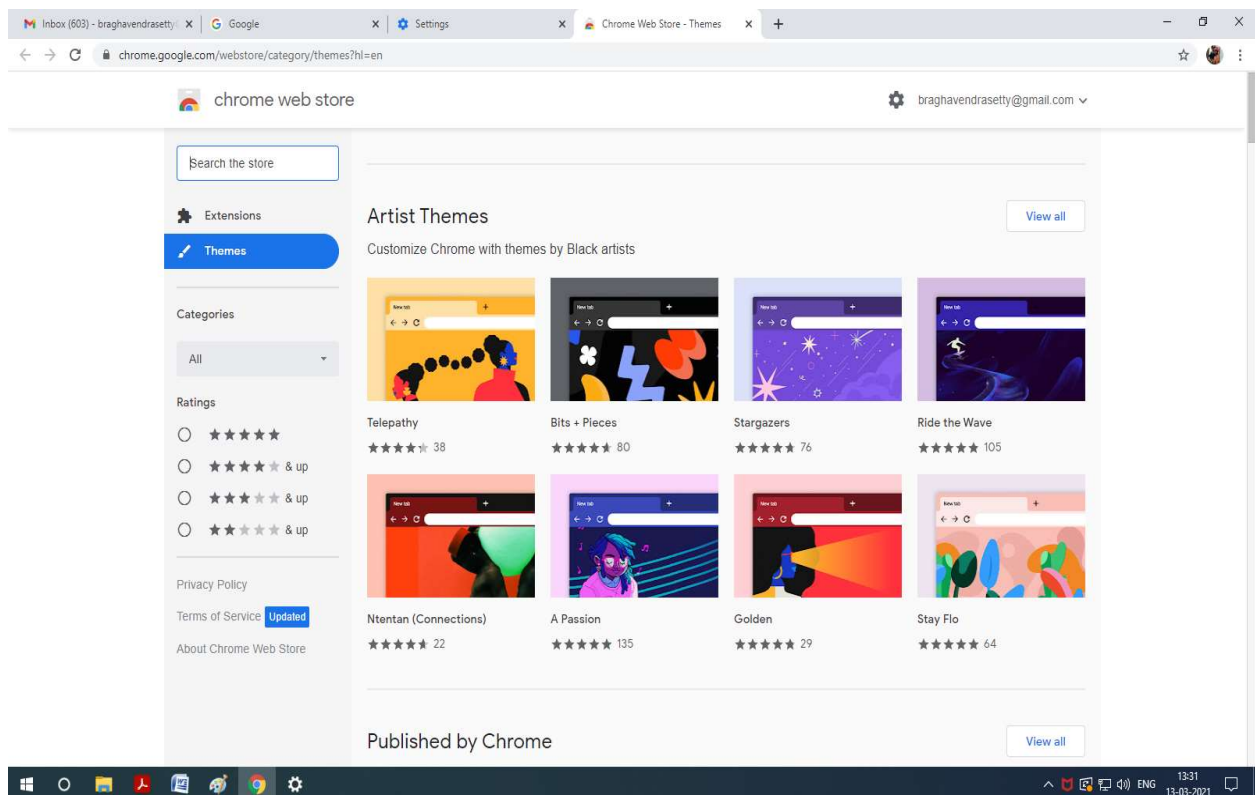
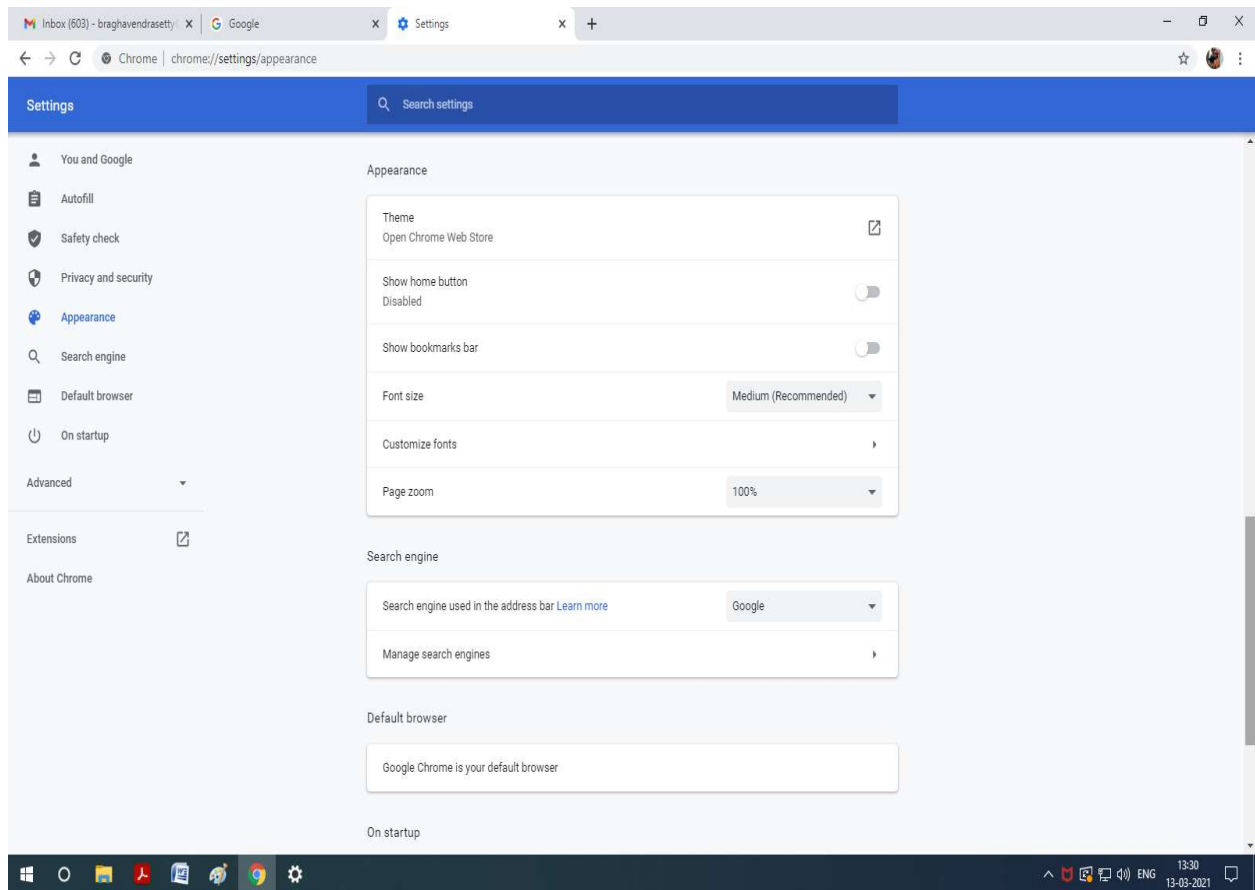
Step 7: In this setting we can change “Browser Theme”, “Font size”, “Home Buttons”, “Page zoom” etc.

Step 8: Similarly click on other options to make necessary changes.

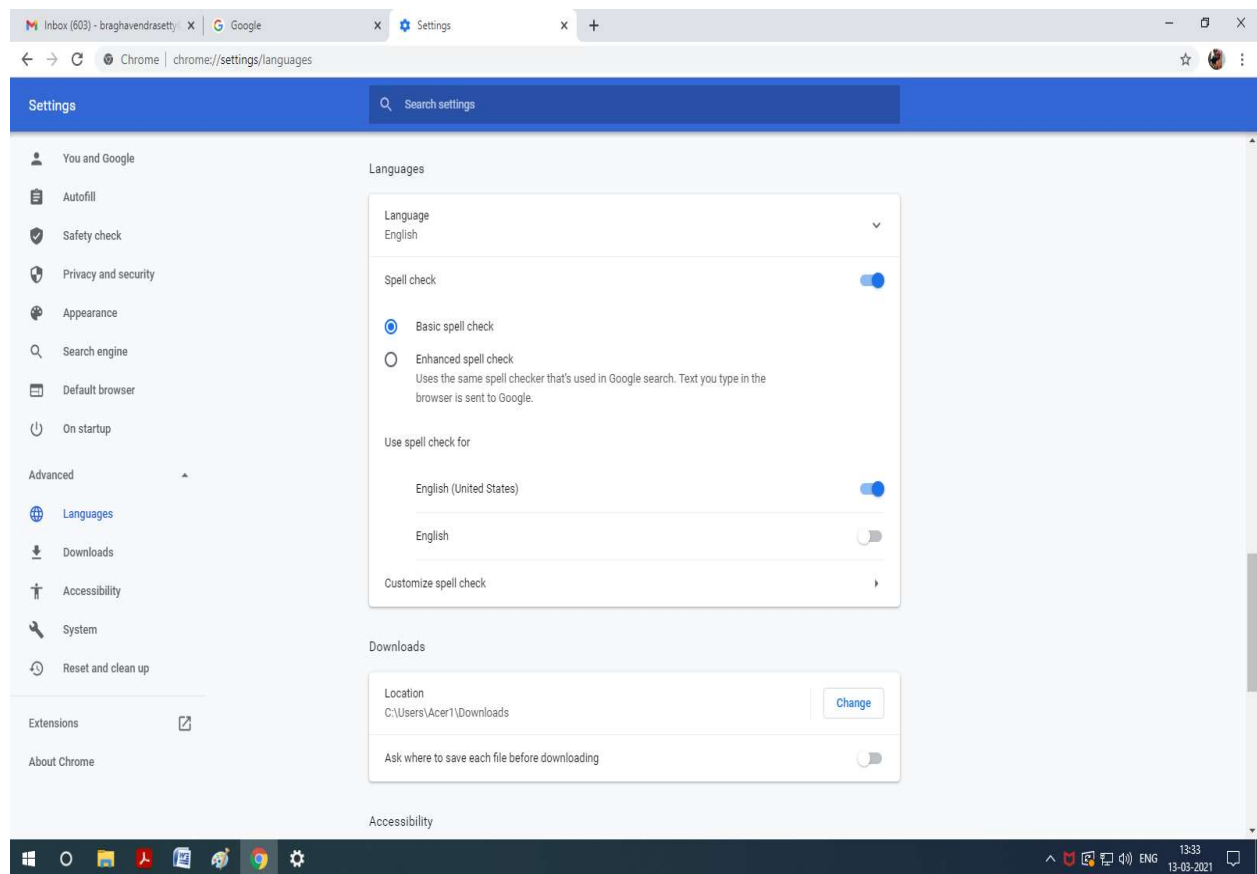
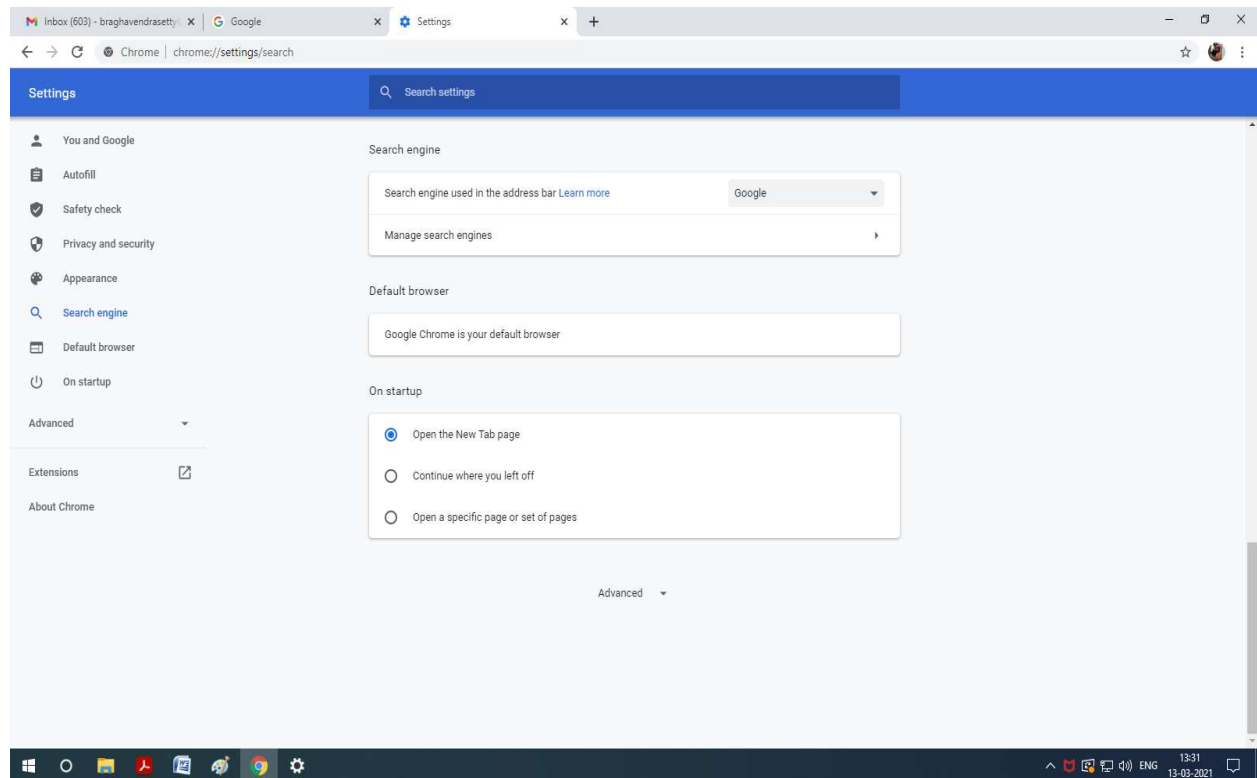
DEMONSTRATION LINKS (YOUTUBE VIDEOS):

Browser Settings in Chrome: <https://www.youtube.com/watch?v=9IIMSzrjUrU>





IT SKILLS LAB MANUAL [20CS01P]



Program 18: Demonstration and hands on privacy settings and password policy “GOOGLE CHROME” PRIVACY SETTINGS

Step 1: Open Google chrome browser.

Step 2: Click Menu bar in the top right corner of the screen.

Step 3: Select settings from the dropdown menu.

Step 4: Here different settings options of browser appears

(Eg. Appearance, Privacy & Security, Search Engine etc.)

Step 5: Click on Privacy & Security.

Step 6: Under "Privacy and security," choose what settings you want to Turn ON and OFF.

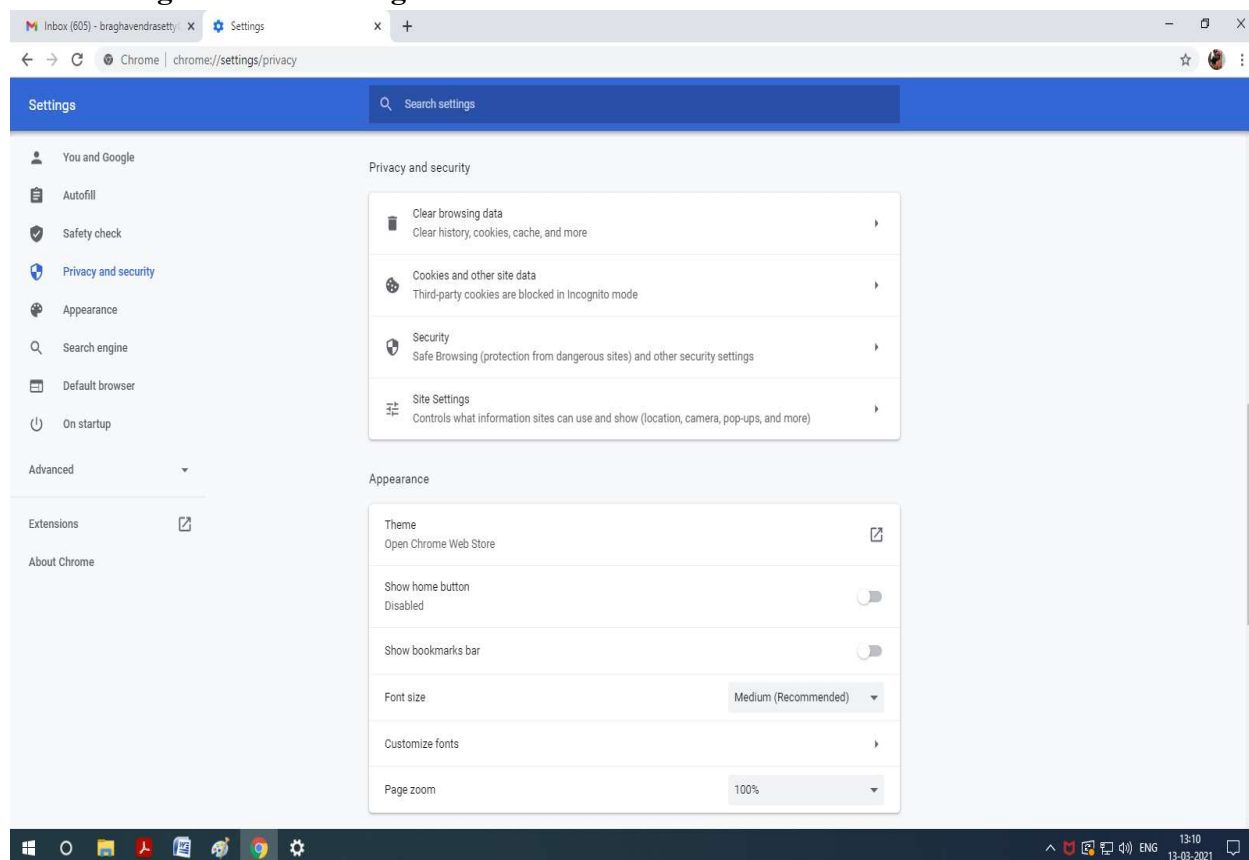
Step 7: On clicking “Clear browsing data” we can delete information from our browsing activity, like our history, cookies, or saved passwords.

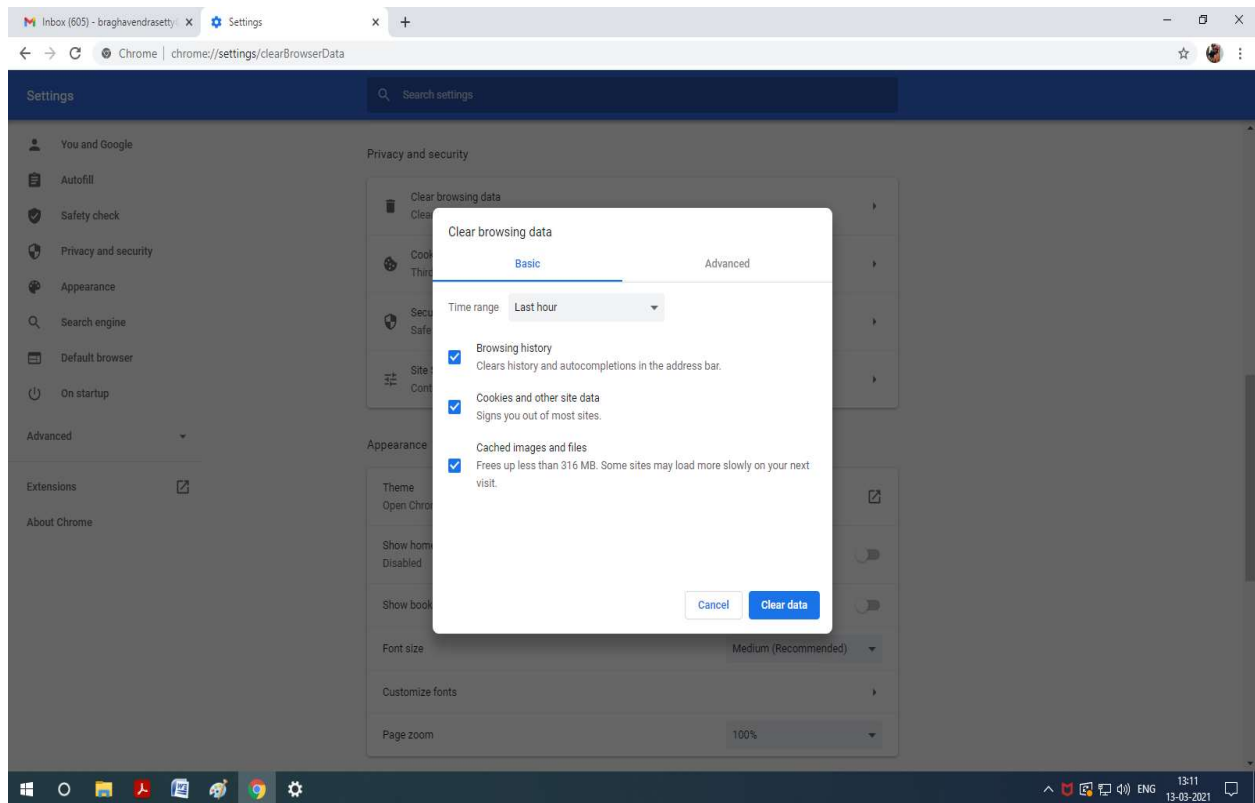
Step 8: On clicking “Cookies and other site data” we can control how Chrome handles cookies and tracking.

Step 9: On clicking “Security” we can manage safe browsing and protection.

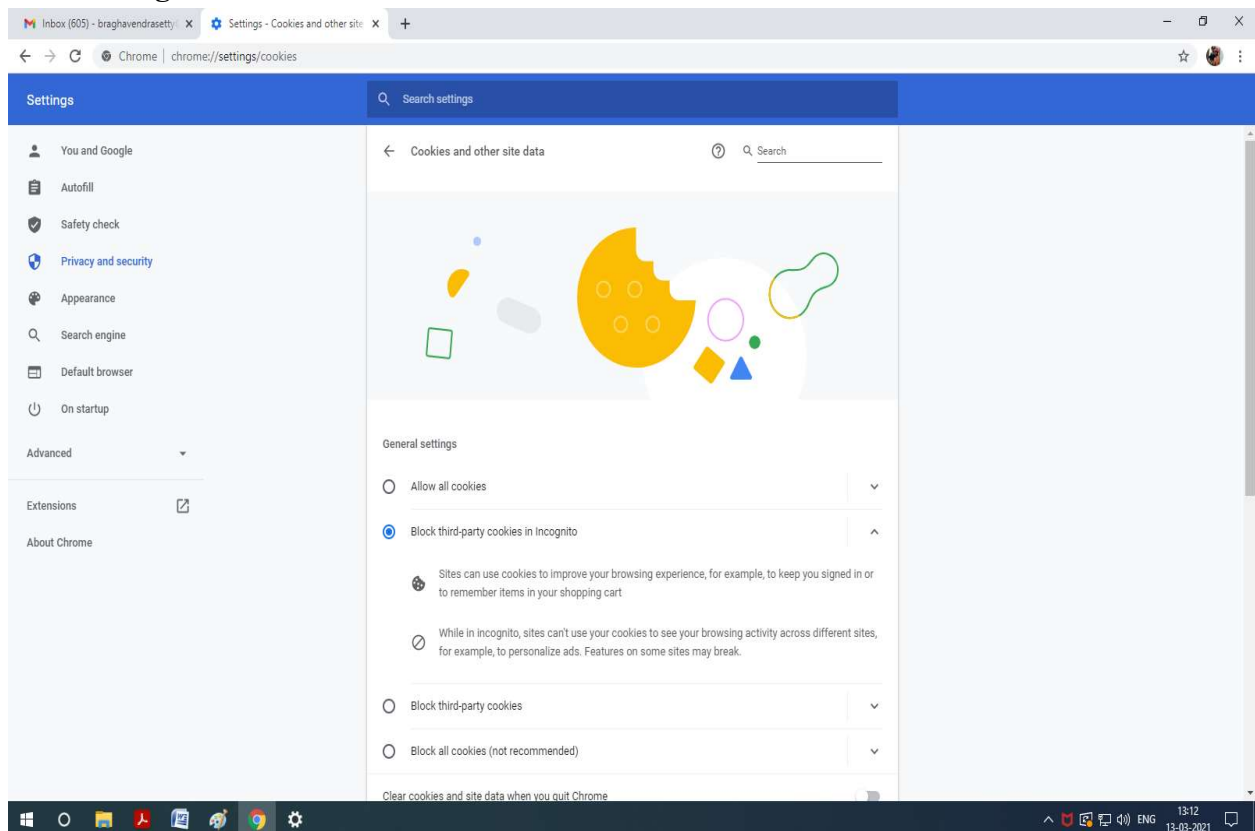
Step 10: On clicking “Site Settings” we can control how Chrome handles content and permissions for a site.

On Clicking “Clear browsing data”

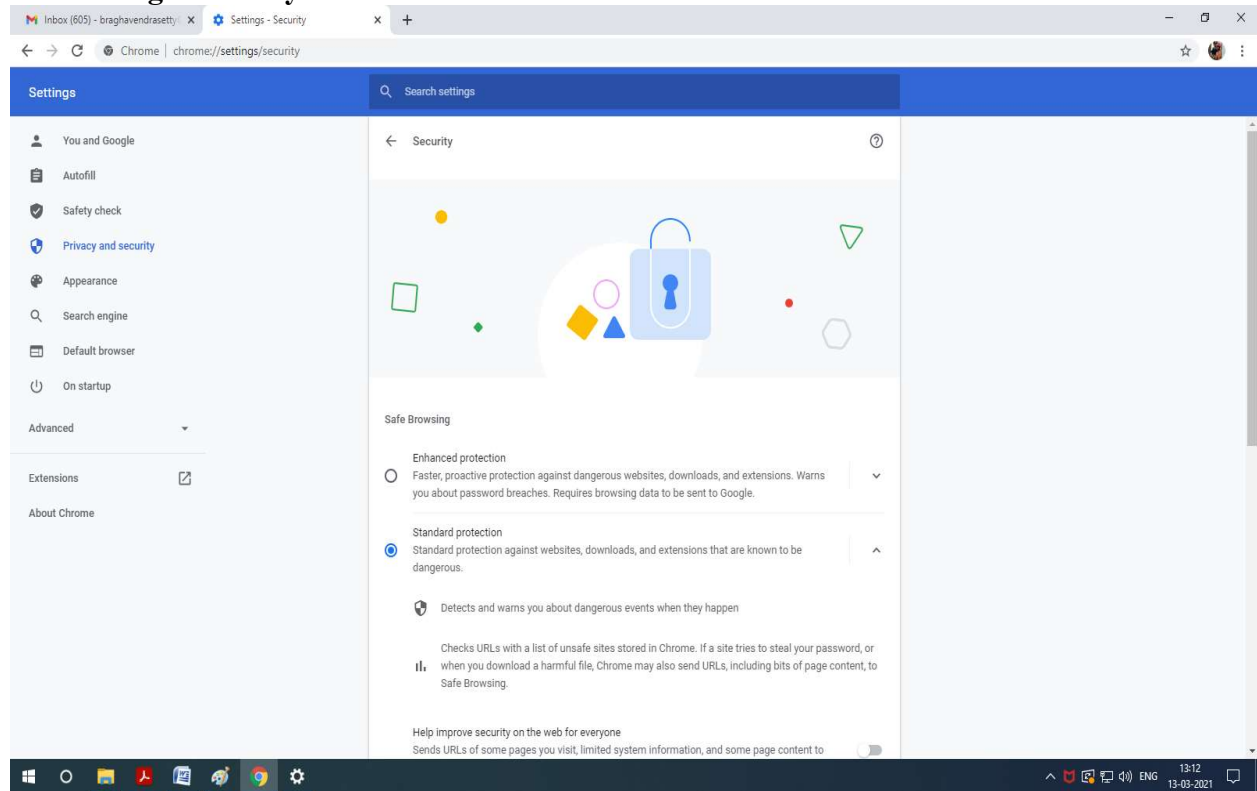




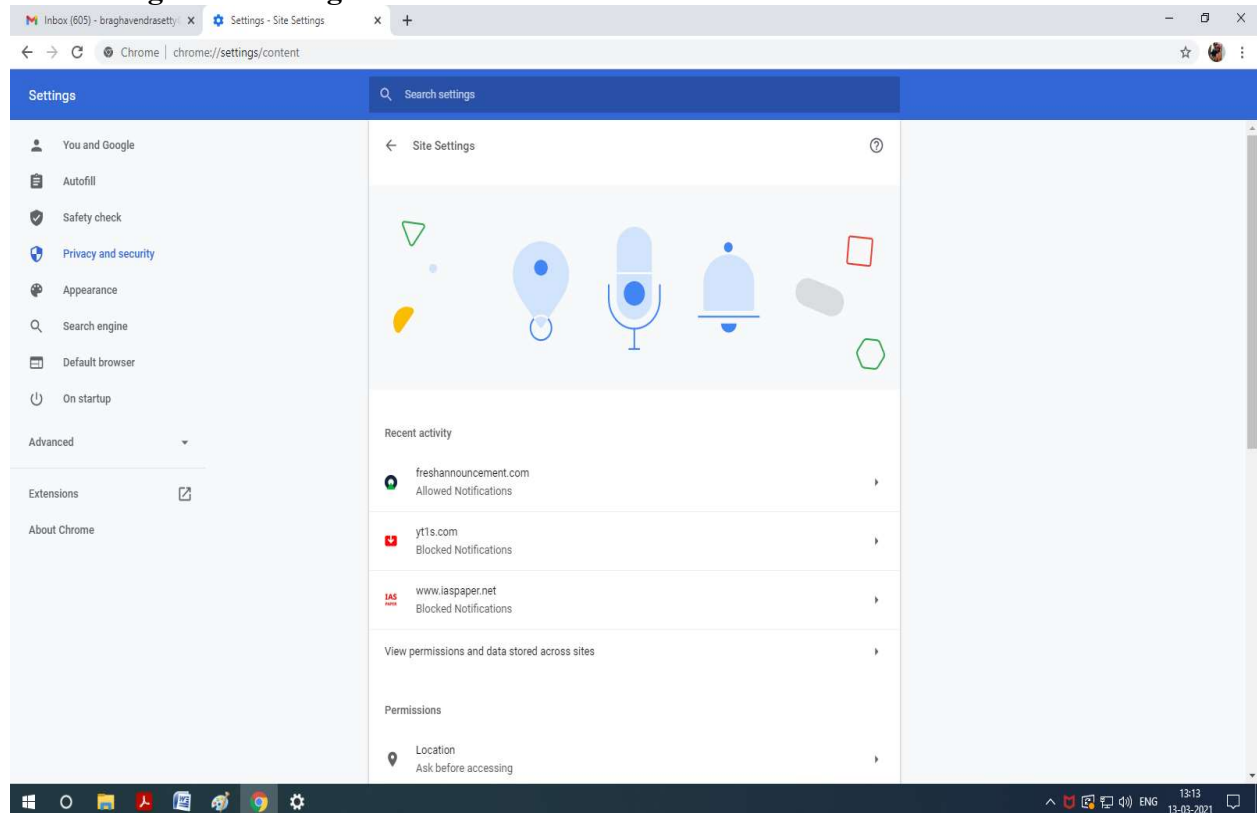
On Clicking “Cookies and other site data”



On clicking “Security”



On clicking “Site Settings”



“GOOGLE CHROME” PASSWORD SETTINGS

Step 1: Open Google chrome browser.

Step 2: Click Menu bar in the top right corner of the screen.

Step 3: Select settings from the dropdown menu.

Step 4: Here different settings options of browser appears

(Eg. Autofill, Appearance, Privacy & Security, Search Engine etc.)

Step 5: Click on “Autofill”

Step 6: Under "Autofill" choose what settings you want to Turn ON and OFF.

Step 7: By enabling “Offer to save passwords” browser will ask to save your password before login and vice versa.

Step 8: By enabling “Auto Sign-in” browser will automatically sign in to websites using stored credentials.

Step 9: On clicking “Check passwords” will keep your passwords safe from data breaches and other security issues.

Step 10: We can “View and manage saved passwords of our Google Account”.

DEMONSTRATION LINKS (YOUTUBE VIDEOS):

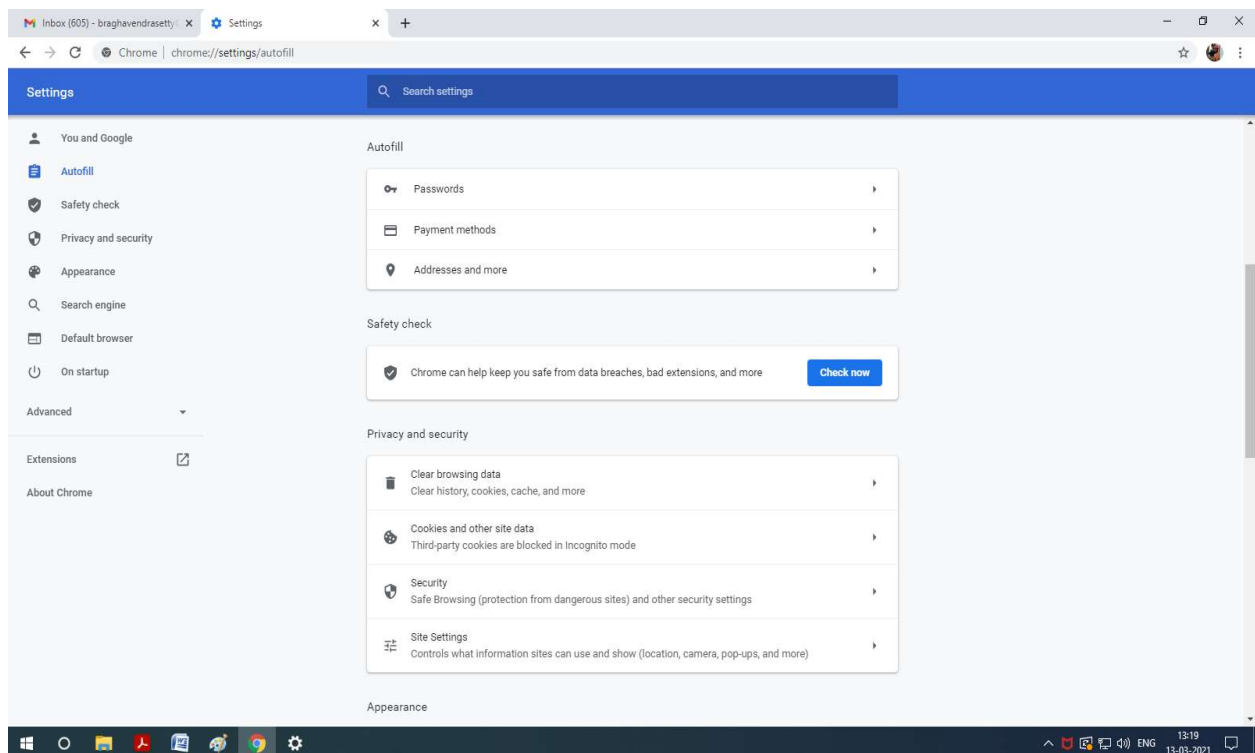
1) Privacy Setting in Chrome:

<https://www.youtube.com/watch?v=zMXI6waGFp4>

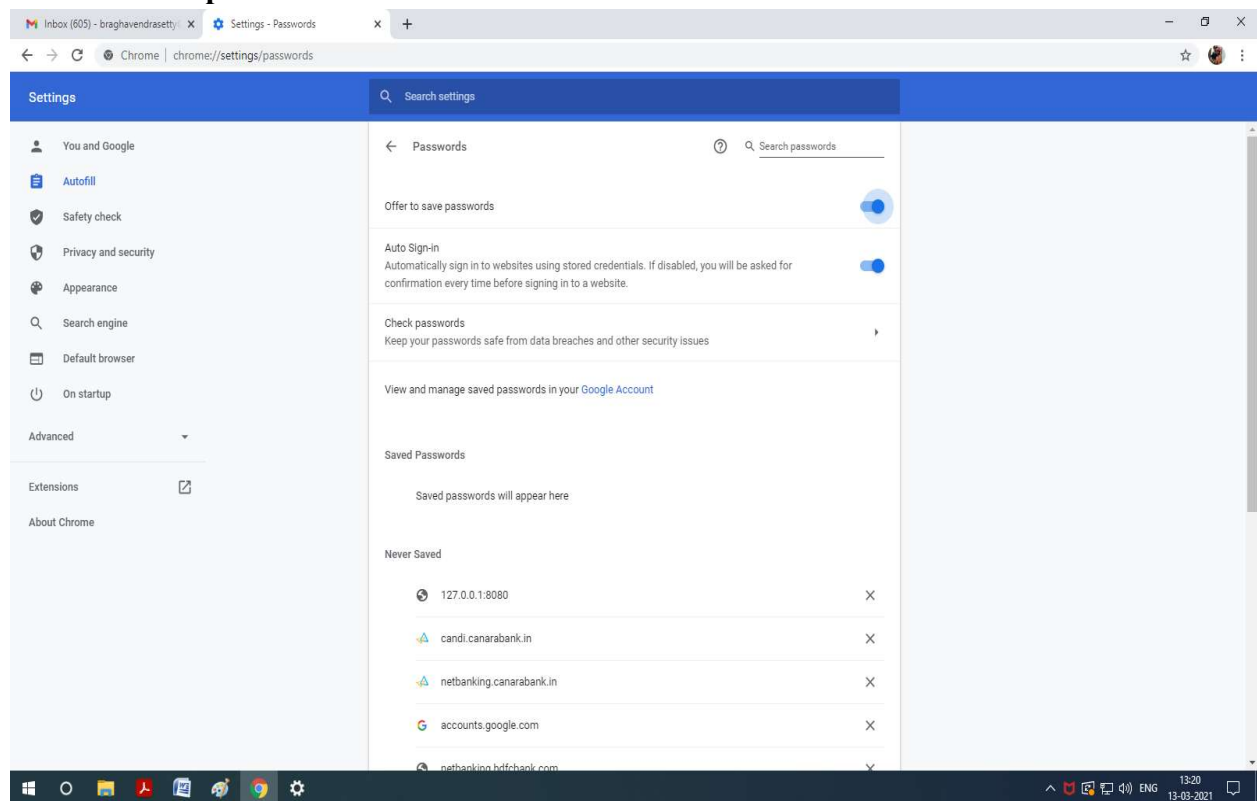
2) Password Setting Policy in Chrome:

<https://www.youtube.com/watch?v=pT1Vcq9AVCo>

"Autofill"



“Offer to save passwords”



Program 19: Demonstration of common security threats (using videos)

- a. Phishing**
- b. DoS attack**
- c. Man in the middle attack**
- d. Spamming**
- e. Virus**

COMPUTER SECURITY

- Computer Security is the process of detecting and preventing any unauthorized use of your laptop/computer. It involves the process of safeguarding against trespassers from using your personal or office based computer resources with malicious intent or for their own gains, or even for gaining any access to them accidentally.
- Cyberspace (internet, work environment, intranet) is becoming a dangerous place for all organizations and individuals to protect their sensitive data or reputation. This is because of the numerous people and machines accessing it.



COMMON SECURITY THREATS

- a) Phishing**
- b) DoS attack**
- c) Man in the middle attack**
- d) Spamming**
- e) Virus**

A) PHISHING:

Phishing is an online method of criminals trying to steal your personal information like login credentials, credits and numbers using deceptive e-mails and numbers.

COMMON “PHISHING” ATTACKS

Email phishing: In this type of phishing the fraudster sends email or links to hundreds or thousands of people to steal their personal data or information.

Spear Phishing: In this type of phishing the fraudster sends email or link to a specific person or organization from a trustworthy source to steal the personal data.

CEO fraud: In this type of phishing the fraudster spoof company emails accounts to send email or links to their employees to steal their personal data.

Vishing: In this type of phishing the fraudster makes a voice call, in order to induce users to reveal personal information by claiming to be from a reputable company.

Smashing: In this type of phishing the fraudster sends a message, in order to induce users to reveal personal information by claiming to be from a reputable company.

Pharming: In this type of phishing the fraudster installs malicious code in your computer which automatically redirects you to fake websites without your knowledge.

HOW TO PREVENT PHISHING

1. Two factor authentication (2FA).
2. Strict password management policies.
3. Not clicking on external e-mail links.

B) DOS (DENIAL OF SERVICES)/DDOS (DISTRIBUTED DOS)

DoS/DDoS is an online method of criminals trying to shut down a machine or network & making it inaccessible to the user.

COMMON “DoS” ATTACKS

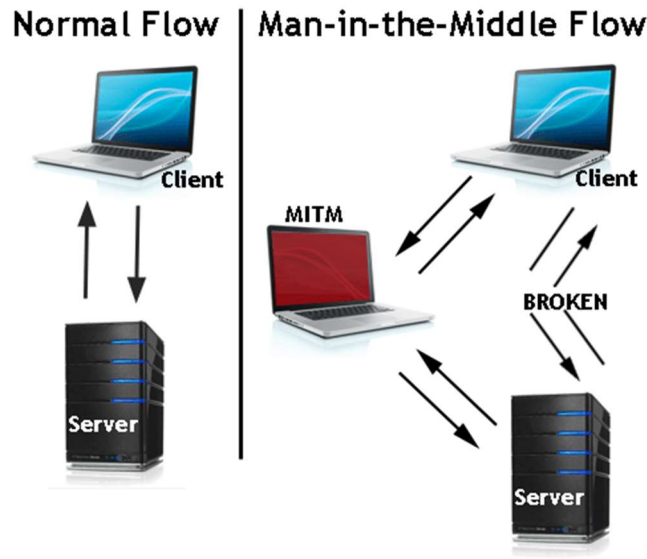
- Volume band attack: In this type of attack the fraudster tries to saturate the bandwidth of the target site, creates traffic jam which makes it impossible to user data to reach its target site (some times in excess of 100gbps). It includes TCP floods, UDP floods, ICMP flood.
- Protocol attack: In this type of attack the fraudster tries to consume network resources like servers, firewall, load balances by targeting protocols (layer 3 & 4) with malicious connection requests.
- Application layer attack: In this type of attack the fraudster tries to attack the application itself focusing on specific issues, resulting in the application not being able to work properly.

PREVENTIONS OF “DoS” ATTACK

1. Secure your network infrastructure.
2. Practice basic network security.
3. Understanding the warning signs.
4. Leverage the cloud.

C) MAN IN THE MIDDLE ATTACK

It is a cyber attack where the attacker secretly alters the communications between two parties who believe that they are directly communicating with each other.



COMMON TECHNIQUES OF MAN IN THE MIDDLE

Sniffing: In this attack the attackers use packet capture tool to inspect packets at low level using specific wireless devices.

Packet Injection: In this attack the attackers injects malicious packets into data communication stream, so that is get mixed into valid (original) data.

Session Hijacking: In this attack the attackers corrupt valid computer session to gain unauthorized access to information or services in a computer system.

Email Hijacking: In this attack the attackers or hackers use tactic to target email accounts of large organization

PREVENTIONS OF “MAN IN THE MIDDLE” ATTACK

1. Use of Virtual Private Network (VPN).
2. Network security.
3. Install active virus and malware protection.
4. Secure your communication.

D) SPAMMING

Spamming is an online method of criminals trying to send unwanted message (spam) to large number of users for the purpose of commercial or any other prohibited purposes.

COMMON “SPAMMING” ATTACKS

- **Ads:** In this method the fraudster send emails / message offering product and services such as weight loss pills and fairness cream etc.
- **Chain letters:** In this method the fraudster send emails / messages telling exiting stories and to pass the same message to 100 or more people giving intuitions.
- **Money scam:** In this method the fraudster send emails / message asking small amount money to receive big rewards in future.
- **Porn spam:** In this method the fraudster sends pornography emails / message using erotic images and videos.
- **Email spoofing:** In this method the fraudster uses email phishing attack methods.

PREVENTION OF “SPAMMING” ATTACK

1. Never post your email address publically.
2. Think before you click on the link.
3. Do not reply to spam messages.
4. Avoid using your personal or business email address for registering online contest.

E) MALWARE (MALICIOUS SOFTWARE)

Malware is a computer program written with the intent to damage/ disable device /system and network. It is used to steal data, bypass access control and harm computers and network devices they are connected.

COMMON TYPES OF “MALWAREs”

1. Virus:

Viruses are designed to damage the target computer by corrupting data, reformatting your hard disk or completely shutting down the system. The full form of virus is “Vital Information Resources under Seize”, they can also be used to steal information, steal many render advertisements and more.

Types of viruses i) Boot virus ii) Program file virus.

2. Worm:

Worms are designed to replicate itself on the target computer and to infect other computers without requiring action from anyone.

3. Trojan Horse:

Trojan Horse are designed to trick you into downloading and installing malware, as soon as you install Trojan, the criminals access your system and can steal personal information and can install more malware to modify files and to destroy your data.

4. Spyware:

Spyware are designed by the criminals to track your browsing habits and internet activity, spying capabilities can include activities like monitoring collecting login credentials, financial information and many more.

5. Ransom ware:

Ransom wares are designed by the criminals that hold your data and demands payment to release the data back to you. It restricts user access to computer by either encrypting files on the hard disk or shutting down the system until the criminal is paid by the user.

DEMONSTRATION LINKS (YOUTUBE VIDEOS):

- 1) Computer Security: <https://www.youtube.com/watch?v=d30n-YxOHo4>
- 2) Phishing attack: <https://www.youtube.com/watch?v=PR0c-gJ20kA>
- 3) DoS attack: <https://www.youtube.com/watch?v=PTJ6UZz1pPQ>
- 4) Man in the Middle attack: <https://www.youtube.com/watch?v=dEaDgOZY80Y>
- 5) Spamming attack: <https://www.youtube.com/watch?v=B47mwZAJXls>
- 6) Virus attack: <https://www.youtube.com/watch?v=n8mbzU0X2nQ>